Mode coupling implementation

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January 2025

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1 Introduction

1.1 The coupling matrices of Al-Attar et al. [2]

This is a working document that discusses the implementation of our new normal mode coupling calculations. In particular, this is really a recipe book which lays out in full detail how to calculate the coupling matrices of Al-Attar et al. [2]. Now, in this paper they derive exact coupling matrices for an aspherical, fluid/solid planet. The expansion of the wavefield is

$$\mathbf{u}(\mathbf{x}, t) = \sum_n u_n(t)\hat{\mathbf{u}}_n(\mathbf{x}), \tag{1}$$

where $\hat{\mathbf{u}}_n = \mathcal{D}\Phi_1 \cdot \mathbf{u}_n$, and Φ_1 is the equilibrium mapping from the referential planet to the physical planet, whilst $\mathbf{u}_n$ is the n-th normal mode (obviously the ordering method by which this is the n-th is irrelevant). The matrices are given as

$$M_{n'n} = \int_M \rho \left\langle \hat{\mathbf{u}}_n, \hat{\mathbf{u}}_{n'} \right\rangle \mathrm{d}^3\mathbf{x}, \tag{2}$$

$$W_{n'n} = \int_M 2\rho \left\langle \boldsymbol{\Omega} \times \hat{\mathbf{u}}_n, \hat{\mathbf{u}}_{n'} \right\rangle \mathrm{d}^3\mathbf{x}, \tag{3}$$

$$H_{n'n} = \int_M \left\{ \rho \left\langle \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \hat{\mathbf{u}}_n) - \gamma_n, \hat{\mathbf{u}}_{n'} \right\rangle + \left\langle \boldsymbol{\Lambda} \cdot \boldsymbol{\mathcal{D}}\hat{\mathbf{u}}_{\boldsymbol{n}}, \boldsymbol{\mathcal{D}}\hat{\mathbf{u}}_{\boldsymbol{n'}} \right\rangle \right\} \mathrm{d}^3\mathbf{x} + \ldots, \tag{4}$$

and the equation of motion is

$$M \cdot \ddot{u} + W \cdot \dot{u} + Hu = f, \tag{5}$$

where u is the vector of expansion coefficients.

1.2 A short digression

In preparation for a more in-depth exploration of the coupling matrices we must first discuss a few points on the solution of the equation:

1. The techniques used to solve the equation are most easily applied when considering the complex spherical harmonics. Consequently it is worthwhile setting up the problem to most easily cope with complex eigenfunctions. The method by which this is achieved is the use of the complex conjugate when considering the test function.

2. The equation of motion is best expressed with operators, in the frequency domain, using bra-ket notation, as

$$\omega^2 \mathcal{M} \left| u \right\rangle + i\omega \mathcal{W} \left| u \right\rangle + \mathcal{H} \left| u \right\rangle = \left| f \right\rangle. \tag{6}$$

   We use $\langle n|$ to denote the bra-vector associated with the n-th normal mode. The basis functions that we use for the expansion are the transformed normal modes, ie $\langle \hat{n}| = \mathcal{B} \langle n|$. We use these to write the equation of motion as

$$\omega^2 \left\langle \hat{n} \right| \mathcal{M} \left| u \right\rangle + i\omega \left\langle \hat{n} \right| \mathcal{W} \left| u \right\rangle + \left\langle \hat{n} \right| \mathcal{H} \left| u \right\rangle = \left\langle \hat{n} \right| \left| f \right\rangle. \tag{7}$$

   Now, when we perform the inverse Fourier-Laplace transform we need to find the matrix inverse $S(\omega)$. However, Al-Attar, Woodhouse, and Deuss [1] showed that it was more efficient (and indeed probably more accurate) to use iterative methods to find the approximate solution to the matrix-vector problem. There is a significant cost associated with defining the matrices M, W, and H. Consequently, it may be faster to never actually define them, but to use the operators themselves in the iterative scheme.

3. With reasonably straightforward pre-conditioning (diagonal matrix with 1D eigenfrequencies) we require generally $\mathcal{O}(1)$ iteration. We express the cost of a single matrix-vector product evaluation as C_1. The total cost of solution when the matrices are never defined is then $N_F C_1$, where N_F is the number of frequencies at which we evaluate the solution. The cost of evaluating the matrices explicitly is $N_M^2 C_1$, where N_M is the number of modes. The cost of an explicit matrix-vector evaluation is N_M^2 approximately. Consequently, if $N_M^2 > N_F$, it becomes more efficient to never compute the matrices. As N_F rarely exceeds 100,000, and N_M can easily be 1000 it most likely will be more efficient to never find M, W, H.

1.3 Normal modes of a 1D Earth

1.3.1 Characterisation

The normal modes of a one dimensional Earth are distinguished by

1. Their type, ie whether they are Toroidal or Spheroidal. These two distinct types of modes decouple from one another in the equations of motion, when the wavefield is expressed in terms of vector spherical harmonics. Toroidal (or "twisting") modes are characterised by having tangential displacements with zero divergence, leaving the shape and density of the one-dimensional Earth the same. Spheroidal modes, in contrast, have radial expression and do not leave the density unaltered.

2. Overtone number. The overtone number indicates different solutions of the radial integral equations.

3. Order and degree. These correspond to the order and degree of the vector spherical harmonics (or indeed generalised spherical harmoncis) used to expand the modes.

1.3.2 Expansion

Although traditionally the vector spherical harmonics were often used, it makes the most sense to instead use the generalised spherical harmonics. In this basis set we have the expansion

$$\mathbf{u} = u^-(r) Y_{lm}^-(\theta, \varphi)\hat{\mathbf{e}}_- + u^0(r) Y_{lm}^0(\theta, \varphi)\hat{\mathbf{e}}_0 + u^+(r) Y_{lm}^+(\theta, \varphi)\hat{\mathbf{e}}_+. \tag{8}$$

The equations of motion separate and one obtains three radial, coupled, differential equations for u^-, u^0 and u^+. In terms of the coefficients that one obtains from doing the same, but with vector spherical harmonics, the result is

$$U = u^0, \qquad V = \frac{1}{\sqrt{2}}(u^- + u^+), \qquad W = \frac{i}{\sqrt{2}}(u^- - u^+). \tag{9}$$

Consequently one can use the output of a program like MINEOS to obtain the expansion coefficients with the result

$$u^0 = U, \qquad u^+ = \frac{1}{\sqrt{2}}(V + iW), \qquad u^- = \frac{1}{\sqrt{2}}(V - iW). \tag{10}$$

1.3.3 Important points

One of the key properties of the radial differential equations for the various components is that they are independent of the order m. Consequently, storing the normal modes of a one-dimensional Earth

is computationally cheap. In particular, one only needs to store u^{α}_{lm} once for each l. This property, however, breaks down in a three-dimensional Earth. The basis functions of the expansion that we use, however, are not the normal modes themselves. They are, instead $B \cdot u_n$. Consequently, they differ for every m. Storing these is computationally prohibitive (or indeed probably not possible for a large calculation), given that the scaling of the number of modes is l^2 and the results would need to be stored on the numerical grid used for the spectral element scheme. It is, therefore, worthwhile considering all calculations as requiring the use of the one-dimensional normal modes rather than the basis functions themselves. Although this adds slightly to the computational cost of the calculations, the benefit in terms of storage is significantly bigger.

It is important to note, as well, that each normal mode simply has expression in generalised spherical harmonics at a single l, m. This is very useful for the purpose of generalised spherical harmonic transforms.

1.4 The transformation of an arbitrary vector

An operation that will come up frequently is integrating, over the referential model, a particular normal mode, against an arbitrary vector, i.e.

$$\int_M \overline{\langle n|} \cdot |q\rangle \, \mathrm{d}^3\mathbf{x}. \tag{11}$$

We proceed by expanding $|q\rangle$ in terms of generalised spherical harmonics and writing $\langle n|$ in the long form, i.e.

$$\begin{aligned}
\int_M \overline{\langle n|} \cdot |q\rangle \, \mathrm{d}^3\mathbf{x} &= \int_M \overline{\sum_{\alpha} u^{\alpha}(r) Y^{\alpha}_{lm} \hat{\mathbf{e}}_{\alpha}} \cdot \sum_{l'm'\beta} q^{\beta}_{l'm'}(r) Y^{\beta}_{l'm'} \hat{\mathbf{e}}_{\beta} \, \mathrm{d}^3\mathbf{x}, \\
&= \sum_{l'm'} \int_r \mathrm{d}r r^2 \int_{\Omega} \mathrm{d}\Omega \sum_{\alpha} \overline{u^{\alpha}(r)} q^{\alpha}_{l'm'}(r) \overline{Y^{\alpha}_{lm}} Y^{\alpha}_{l'm'}, \\
&= \sum_{\alpha} \int_r \mathrm{d}r r^2 \overline{u^{\alpha}(r)} q^{\alpha}_{lm}(r),
\end{aligned} \tag{12}$$

where the properties of the canonical basis set and generalised spherical harmonics have been used.

1.5 Cost associated with operations

1.5.1 The grid

To work out the cost associated with the calculation of the matrices, or rather with the evaluation of a matrix vector product, we need to firstly consider the spatial grid at each radius that one has for the GSPH. For a given $l_{\max}$ the grid is $(l_{\max}+1) \times (2l_{\max})$. There are $l_{\max}+1$ colatitudes and $2l_{\max}$ longitude points. The radial structure is N_E radial elements and N_P points within each element.

1.5.2 GSPH

The generalised spherical harmonics are given by $Y_{lm}^N = X_{lm}^N(\theta)e^{im\varphi}$. Consequently, the transform of $f(\theta,\varphi)$ from spatial to GSPH basis is given by

$$\begin{aligned} f_{lm}^N &= \int_{\mathcal{S}^2} \mathrm{d}\Sigma \overline{X_{lm}^N(\theta)e^{im\varphi}} f(\theta,\varphi), \\ &= \int_0^\pi \mathrm{d}\theta \sin\theta \overline{X_{lm}^N} \int_0^{2\pi} \mathrm{d}\varphi e^{-im\varphi} f(\theta,\varphi). \end{aligned} \tag{13}$$

The inner integral is simply a Fourier transform. Using the Fast Fourier Transform algorithm this has a complexity of $n \log n$ where $n = 2l_{\max}$ is the number of points. There are $l_{\max}+1$ points at each longitude and consequently the complexity of the FFTs is $(l_{\max}+1) \times 2l_{\max}\log(2l_{\max})$. Now, for each lm, we need to perform the integration. A single integration is

$$\int_0^\pi \mathrm{d}\theta \sin\theta \overline{X_{lm}^N(\theta)} g(\theta). \tag{14}$$

The colatitude points are chosen to be the Gaussian quadrature points. This means that to integrate we require

$$\int_0^\pi \mathrm{d}\theta \sin\theta \overline{X_{lm}^N(\theta)} g(\theta) = \sum_{n=0}^{l_{\max}} c_n \sin\theta_n \overline{X_{lm}^N(\theta_n)} g(\theta_n). \tag{15}$$

This requires $3(l_{\max}+1)$ complex multiplications and $l_{\max}$ complex additions. Both multiplication and addition are similar speeds (or have the same scaling) and we have $4l_{\max} \sim l_{\max}$ operations per integral. The integral is performed for each lm and consequently there are $(l_{\max}+1)^2$ integrals. The cost of the integrals is, therefore, approximately $l_{\max}^3$. Overall therefore we have an approximate cost per GSPH of $l_{\max}^3 + l_{\max}^2 \log l_{\max}$. Testing this in practice, we find the cost to do 100 transformations for different $l_{\max}$ is given in Table 1. As can be seen in this table the exact scaling is dependent upon some implementation and probably some other details. In calculations of approximate cost we will use $\mathcal{O}(l_{\max}^3)$. Although this may not perfectly capture cost, it is good enough to be a reasonable approximation.

$l_{\max}$	t	Ratio	$\mathcal{O}(l^3)$	$\mathcal{O}(l^3 + l^2 \ln l)$	$\mathcal{O}(l^3 \ln l)$
100	0.102	1.0	1.0	1.0	1.0
200	0.809	7.9	8.0	7.85	9.2
400	6.17	60.5	64	62	83
800	48.1	471	512	494	743

Table 1: Table of times for GSPH for different $l_{\max}$

1.5.3 Transformation of an arbitrary vector

The operation we wish to find the cost of is the operation in (11). We have a vector $|q\rangle$ which we have in spherical harmonic coefficients. We wish to integrate $\int_M \overline{\langle n|} \cdot |q\rangle \, \mathrm{d}^3\mathbf{x}$ for all normal modes $|n\rangle$. We showed that this is equivalent to

$$\int_M \overline{\langle n|} \cdot |q\rangle \, \mathrm{d}^3\mathbf{x} = \sum_\alpha \int_r \mathrm{d}r r^2 \overline{u^\alpha(r)} q_{lm}^\alpha(r). \tag{16}$$

This integral is evaluated by using the GLL points within each radial element and adding up each segment. Just focussing on the scaling we have order N_P operations per element and consequently $N_E N_P$ operations per radial integral. It is difficult to analyse specifically how many operations are required at each radius for the integral, as implementation will vary. Approximately however one could say there is one multiplication for each α and then a sum over the α's, followed by multiplication against $w_n r_n^2$. Thus, there are approximately $8N_E N_P$ operations. It is, however, probably not useful to consider the factor of 8. Now, this operation has to be performed for all modes, consequently the cost is given by $\mathcal{O}(N_E N_P N_M)$, where N_M is the number of modes in the expansion. Now the total number of modes under a given frequency will scale approximately as f^3. As asymptotically the frequency of a mode branch scales as $l_{\max}$, it follows that $N_M \sim l_{\max}^3$. Thus, the overall cost scales approximately as $\mathcal{O}(N_E N_P l_{\max}^3)$.

1.5.4 Evaluation into generalised spherical harmonics

We have an arbitrary vector $|u\rangle = \sum_n u_n |n\rangle$. We would like to have it in terms of spherical harmonics at each radius. Now, we know that

$$|n\rangle \equiv \frac{1}{\sqrt{2}}(V - iW)Y_{lm}^{-}(\theta,\varphi)\hat{\mathbf{e}}_- + U(r)Y_{lm}^{0}(\theta,\varphi)\hat{\mathbf{e}}_0 + \frac{1}{\sqrt{2}}(V + iW)Y_{lm}^{+}(\theta,\varphi)\hat{\mathbf{e}}_+. \tag{17}$$

Now, $l_{\max}$ is chosen such that $l_{\max}$ is the maximum l in all the modes. We also have to sum over all the overtones which are indicated with the subscript k. We therefore have

$$\begin{aligned} |u\rangle &= \sum_n u_n |n\rangle \,, \\ &= \sum_k \sum_{lm} u_{klm} |klm\rangle \,, \\ &= \sum_k \sum_{lm} u_{klm} \left\{ {}_kU_l Y_{lm}^0(\theta,\varphi)\hat{\mathbf{e}}_0 + \frac{1}{\sqrt{2}}({}_kV_l \pm i\,{}_kW_l)Y_{lm}^{\pm}(\theta,\varphi)\hat{\mathbf{e}}_\pm \right\}. \end{aligned} \tag{18}$$

This implies that

$$u_{lm}^0 = \sum_k u_{klm}({}_kU_l), \qquad u_{lm}^{\pm} = \sum_k \frac{1}{\sqrt{2}} u_{klm}({}_kV_l \pm i\,{}_kW_l). \tag{19}$$

The sum over k is required to account for all the overtones. Ultimately, at each radius, one has to perform at least one FPO per mode, so at each radius the cost is $\mathcal{O}(l_{\max}^3)$. Ultimately, therefore, the cost of finding the generalised spherical harmonic expansion at all radii is $\mathcal{O}(N_E N_P l_{\max}^3)$.

1.5.5 Multiplication spatially

One of the key operations that needs to be performed is multiplication of either a scalar or vector field by a tensor field. Obviously, the cost of this will vary with the operation, for instance

1. Multiplying two scalar fields requires one FPO per gridpoint.
2. Multiplying a 3×3 matrix by a vector at all points requires 12 FPO per gridpoint.

However, although the numerical prefactor can vary significantly, the overall scaling is the same regardless of the exact operation. As we have $\mathcal{O}(l_{\max}^2)$ gridpoints at each radius we have a total cost of $\mathcal{O}(N_E N_P l_{\max}^2)$.

1.5.6 Cost discussion

The analysis presented above largely discounted numerical prefactors in the cost of operations. These can be quite significant and affect computational speed quite a lot. Nonetheless, it is often easier to analyse this empirically to assess what parts of the code take the longest. Finally, the scaling seen in actual implementations is often different from that seen in the analysis of the raw algorithms, due to factors, e.g. memory access.

2 The operator $\mathcal{M}$

2.1 Calculation

We wish to calculate the effect of the matrix M on a vector u, which is, equally, given by $\langle \hat{n} | \mathcal{M} | u \rangle$. We firstly need a quick examination of the expansion of the vector $|u\rangle$. In particular, we have

$$\begin{aligned} |u\rangle &= \sum_n u_n |\hat{n}\rangle , \\ &= \sum_n u_n \mathcal{B} |n\rangle , \\ &= \mathcal{B} \sum_n u_n |n\rangle . \end{aligned} \tag{20}$$

The result that we need to compute therefore, is

$$\langle \hat{n} | \mathcal{M} | u \rangle = \int_M \rho \overline{\langle n |} \cdot B^\dagger \cdot B \cdot \sum_{n'} u_{n'} |n'\rangle \, \mathrm{d}^3\mathbf{x}. \tag{21}$$

The wavefield expansion coefficients $u_{n'}$ are, by definition, stored in terms of normal modes. However, B will be stored spatially. To perform the multiplication, therefore, it makes sense to multiply spatially. Thus, we first transform from the normal mode expansion to spatial values. To do this we need to get the coefficient of each spherical harmonic at each spatial point. This is achieved using the radial function of each normal mode and evaluating the coefficient for each generalised spherical harmonic. Then, one can multiply through by $\rho B^\dagger B$ forming $|q\rangle$ spatially. One then transforms $|q\rangle$ to the generalised spherical harmonic basis. Finally we have the result from the previous section and the integral is complete.

The process is summarised by:

1. Transform $|w\rangle = \sum_n u_n |n\rangle$ into the spatial basis. This is achieved by a single generalised spherical harmonic transform.
2. Form $|q\rangle = \rho B^\dagger B |w\rangle$. This is achieved with matrix multiplication at all points on the numerical grid. See the section on matrix multiplication with generalised spherical harmonic expansions.
3. Transform $|q\rangle$ into the generalised spherical harmonic basis at all radial nodes.
4. Perform the radial integrals given in the previous section.

A schematic detailing the steps required is given in Fig. 1. It can be seen that the overall cost (ignoring prefactors) of this operation is $\mathcal{O}(N_E N_P l_{\max}^3)$. It will be interesting to see the scaling when implemented. Nonetheless, it is not really worth doing a more detailed analysis of the prefactors, because there is no obvious way to evaluate $M |w\rangle$. This situation changes with other matrix vector products.

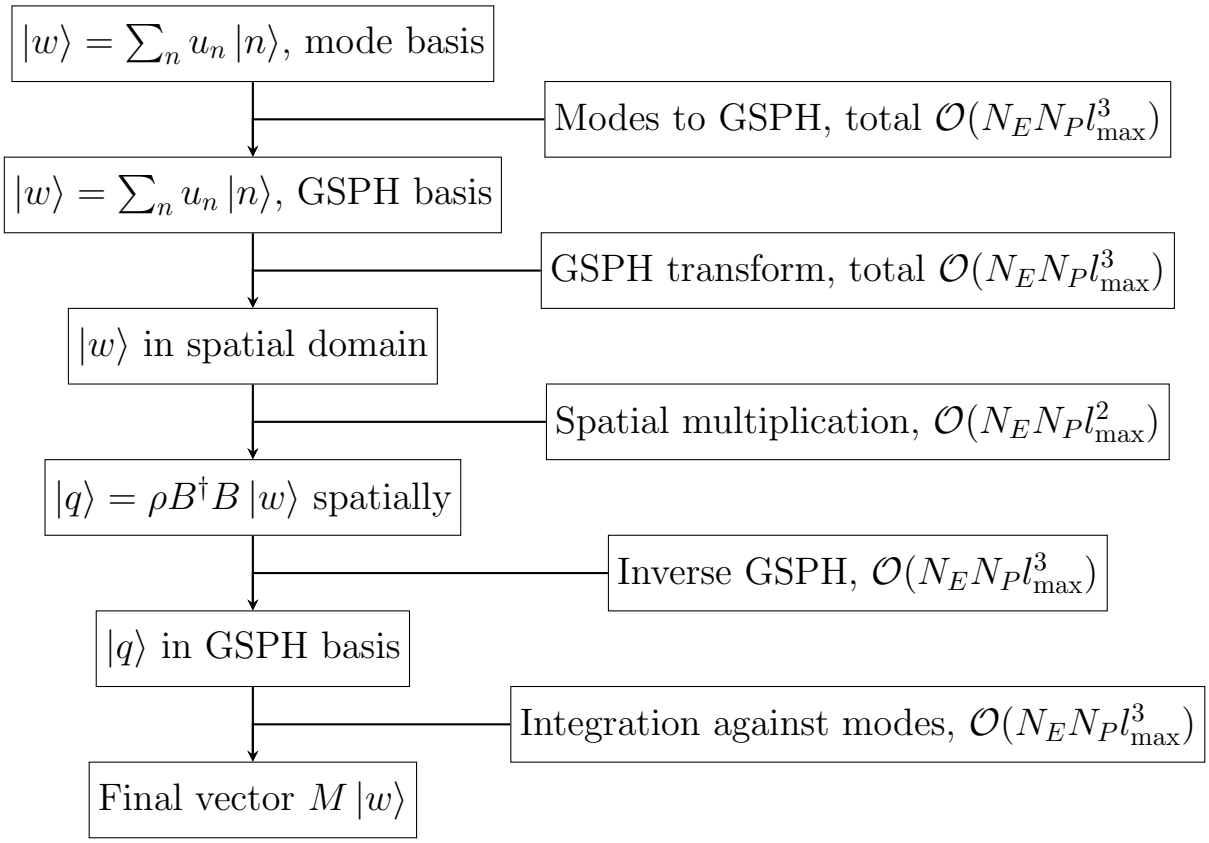

Figure 1: A flowchart depicting, schematically, an inverse problem

3 The operator $\mathcal{W}$

We have

$$W_{n'n} = \int_M 2\rho \left\langle \mathbf{\Omega} \times \hat{\mathbf{u}}_n, \hat{\mathbf{u}}_{n'} \right\rangle \mathrm{d}^3\mathbf{x}. \tag{22}$$

In terms of bra-ket notation we have

$$\langle \hat{n} | \mathcal{W} | u \rangle = 2 \int_M \rho \overline{\langle n |} \cdot B^\dagger \cdot \hat{\Omega} \cdot B \cdot \sum_{n'} u_{n'} | n' \rangle \, \mathrm{d}^3\mathbf{x}, \tag{23}$$

where $\hat{\Omega}$ is the unique matrix associated with the rotation cross-product $\mathbf{\Omega} \times \hat{\mathbf{u}}_n$. To find the resultant vector associated with $W | w \rangle$ the steps are

1. Transform $| w \rangle = \sum_n u_n | n \rangle$ into the spatial basis. This is achieved by a single generalised spherical harmonic transform.
2. Form $| q \rangle = \rho B^\dagger \hat{\Omega} B | w \rangle$. This is achieved with matrix multiplication at all points on the numerical grid. See the section on matrix multiplication with generalised spherical harmonic expansions.
3. Transform $| q \rangle$ into the generalised spherical harmonic basis at all radial nodes.
4. Perform the radial integrals given in the previous section.

4 Evaluating $\mathcal{H}$

There are multiple parts to $\mathcal{H}$ which we denote using numeric subscripts.

5 The BC problem

5.1 Quick summary

The boundary condition problem is as follows. The one-dimensional problem is given by

$$\mathcal{S}_0(\mathbf{u}) = f_0 \text{ on } \Omega, \text{ s.t. } \hat{\mathcal{S}}_0(\mathbf{u}) = 0 \text{ on } \partial\Omega. \tag{24}$$

The full problem is

$$\mathcal{S}(\mathbf{u}) = f \text{ on } \Omega, \text{ s.t. } \hat{\mathcal{S}}(\mathbf{u}) = 0 \text{ on } \partial\Omega. \tag{25}$$

We can ignore changes in the shape of Ω as we are using a referential formulation which will map this into the form of the equation. Solutions for (24) will satisfy the BC $\hat{\mathcal{S}}_0(\mathbf{u}) = 0$ on $\partial\Omega$. However, in general (more below) they will not satisfy $\hat{\mathcal{S}}(\mathbf{u}) = 0$ on $\partial\Omega$. Consequently, if the solutions $\hat{\mathbf{u}}_n$ to (24) are used as a basis set for (25), the solution obtained will not, in general, satisfy the BC $\hat{\mathcal{S}}(\mathbf{u}) = 0$ on $\partial\Omega$.

5.2 The 1D problem

5.2.1 Summary of ideas

To illustrate this difficulty we consider a very simple problem, the 1D one. In particular, we have

$$\mathbf{u}_n = U\mathbf{P_{lm}} + \mathbf{VB_{lm}} + \mathbf{WC_{lm}}. \tag{26}$$

Just considering a spheroidal oscillation the traction is given by (with the following as vector spherical harmonic coefficients)

$$R = (\kappa + \frac{4}{3}\mu)\dot{U} + (\kappa - \frac{2}{3}\mu)(2U - kV), \tag{27}$$

$$S = \mu(\hat{V} - r^{-1}V + kr^{-1}U). \tag{28}$$

The free surface boundary condition becomes $R = S = 0$. If we have a solution to the equation of motion satisfying these conditions, then the functions U and V can satisfy perturbed boundary conditions if and only if $\delta\kappa/\kappa = \delta\mu/\mu$. This implies that the traction-free boundary condition is not necessarily satisfied by the solutions obtained from ALL mode coupling codes currently in existence.

5.2.2 Demonstrating this issue

The issue with this whole thing can be illustrated very neatly using the solutions obtained from MINEOS. In particular, I have used isotropic PREM with no ocean. I have used the modes ${}_nS_1$ up to $n = 400$ (as we are using vector spherical harmonics we don't need to include different l values). For the perturbed model I have simply changed the P-wave velocity from 5800 m s^{-1}, to 5900 m s^{-1} in the upper two layers of PREM. I have then performed a mode expansion:

1. Used layers of PREM
2. Taken PREM200 and used cubic spline between all points
3. Problem divided into layers, the elements within layers are given by the nodes of PREM200
4. Used Gaussian quadrature for all integrals
5. MINEOS eigenfunctions satisfy orthonormality up to around 10^{-6}
6. To get accurate result used gram matrix and LU (can do Cholesky but so fast it doesn't matter)

In the following figures we plot the result of this expansion. In particular, we plot a comparison of the perturbed result and the basis expansion, the norm against number of basis functions and the traction.

It can be seen in Fig. 2 that the relative error is at its maximum near the surface. In Fig. 3 we can see that the L^2 norms converge, however the H_0^1 norm does not. Finally in Fig. 5 we see that the traction does not vanish as is clear from the equations of motion. In Fig. 4 we see that the l2norm is monotonically decreasing with increasing number of basis functions, as we expect.

5.2.3 Notes

Short notes on implementation and interesting things:

1. MINEOS outputs at nodes of the input table (185 nodes)

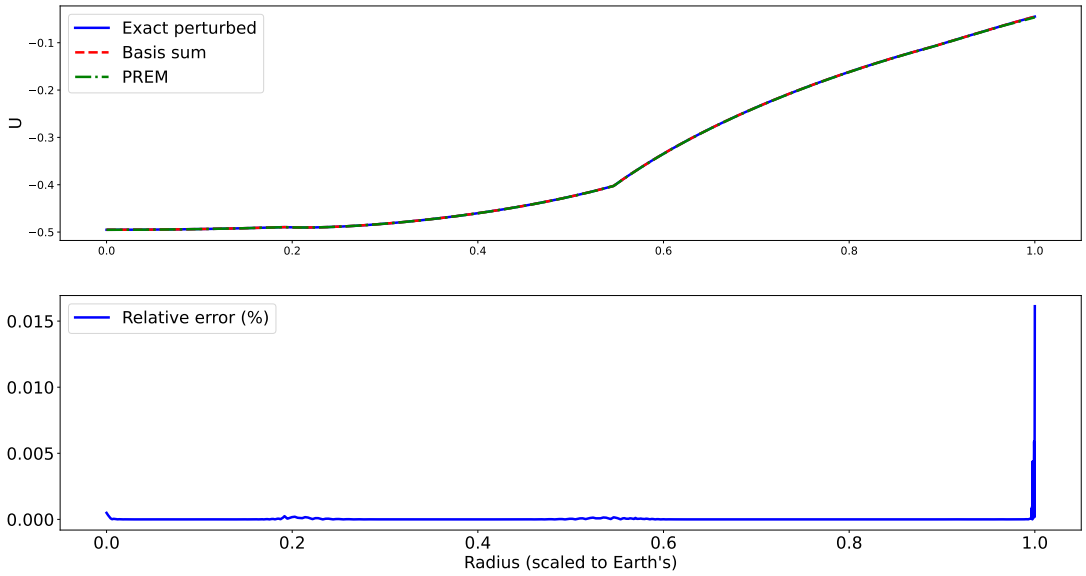

Figure 2: Comparison of perturbed eigenfunction U against the sum from the unperturbed PREM.

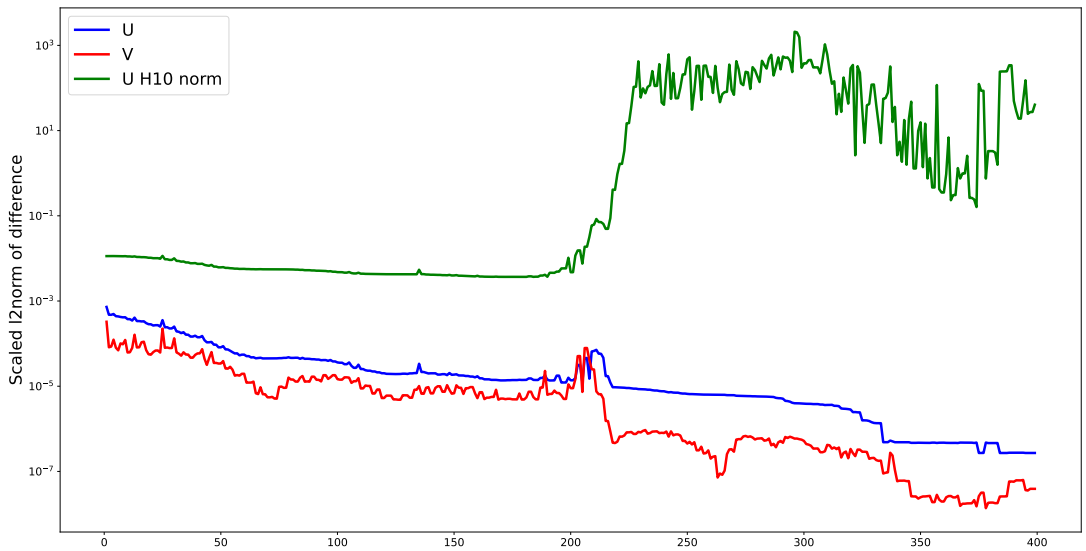

Figure 3: Comparison of norm of error against number of overtones included in the expansion.

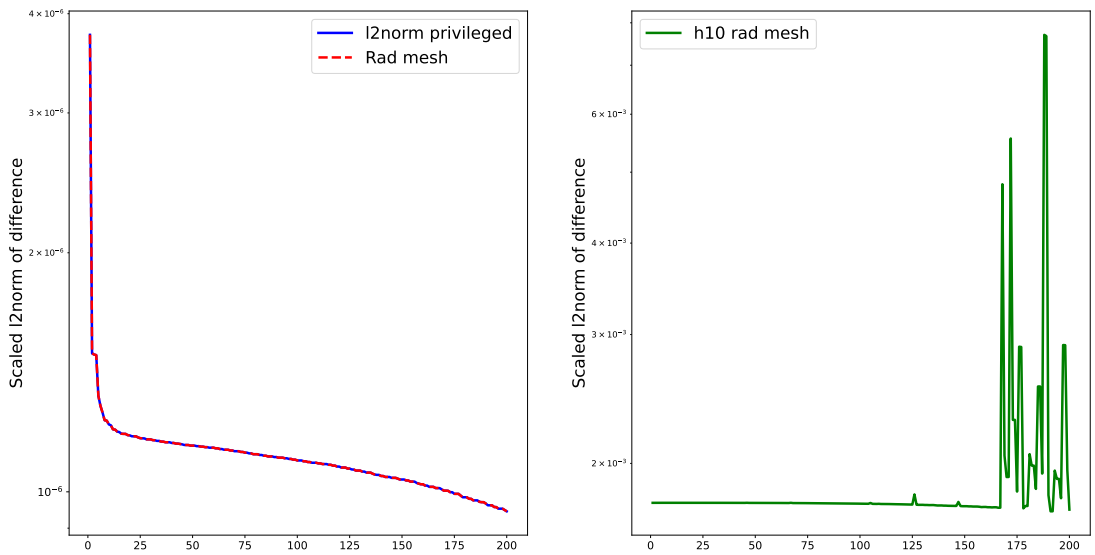

Figure 4: Comparison of l2 norm and h10 norm of error against number of overtones included in the expansion. Also compared are the norms of the radial mesh method and the "privileged" mesh. This demonstrates that despite the issues with the cubic splines interpolation and the resultant integration the results are very similar.

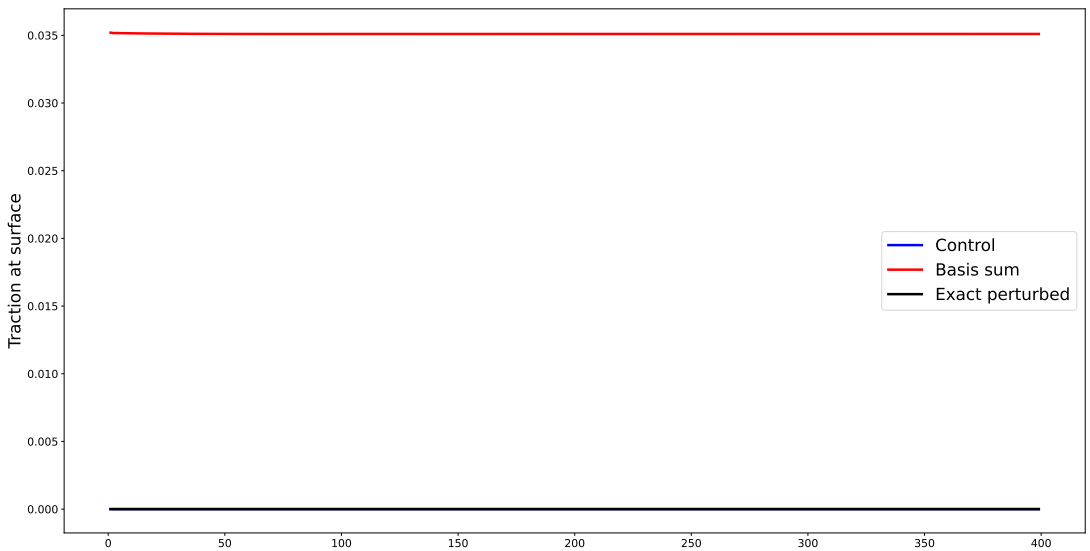

Figure 5: Traction for sum of basis functions in PREM and perturbed PREM.

2. Could try polynomial but haven't
3. Use cubic spline interpolation for output

Initially I just used the output nodes, and then applied Gaussian quadrature between these nodes. This obviously "privileges" the nodes, which is nonsensical. The nodes are just output values. I then switched to using the mesh generated by the radial mesh routine that I used in the gravity calculation. In applying this I found

1. The inner product normalisation was slightly off. Would generally be off by around $1e^{-6}$
2. The "privileged" method and the standard radial mesh resulted in different integrals by a similar amount

To investigate this result I used a cubic splines interpolant on $\sin(x)$ from $0 \to \pi$. With twenty segments the "privileged" approximant had an integral error of around $1e^{-6}$. On the other hand, with only two segments, quadrature had an error of $1e^{-14}$, using the exact values. Applying quadrature with the same segments to the interpolant, the error was similar to the "privileged" one. Consequently it appears that the error we have is probably arising from the splines. One of the things is that the splines obviously are not a continuous higher order polynomial, which perhaps causes issues with the quadrature. Indeed, increasing the order of the quadrature didn't necessarily improve the error against the exact value of the integral when using the interpolant. This indicates that once we have evaluated the eigenfunctions on the grid we are using for the calculation, the numerical values should be regarded as the eigenfunction.

Thoughts:

- Error of IP with "privileged" and standard mesh against exact is basically the same
- If cubic splines interpolant is considered the "exact" then our non-privileged discretisation is giving an error compared to the "exact"
- How does one deal with this though?
- We have to interpolate no matter the output from MINEOS
- If we interpolate with cubic splines then that is our exact result

6 1D analogous problem

In this section I describe an analogous one-dimensional problem which allows us to explore how to resolve the boundary condition issue.

6.1 Setting up the problem

6.1.1 The equation of motion

We look at solving the equation

$$\rho\partial_t^2 u - \partial_x(\mu\partial_x u) = f(x,t), \tag{29}$$

subject to $u(0) = 0$, and $\mu(1)u'(1) + u(1) = 0$. The domain may have layers, with discontinuities in ρ and μ. We impose the requirement on internal discontinuities that $[\mu u' + u]_-^+ = 0$. The initial conditions are $u(0,x) = g(x)$ and $\dot{u}(0,x) = h(x)$.

It is convenient to apply a Laplace transform in the frequency domain, ie taking $\mathcal{L}(f) = \int_0^\infty e^{-i\omega t} f(t)\,\mathrm{d}t$. This yields the equation

$$-\omega^2 \rho \tilde{u}(\omega, x) - \partial_x(\mu \partial_x \tilde{u}(\omega, x)) = \mathrm{i}\omega g(x) + h'(x) + f(\omega, x). \tag{30}$$

If one solves the frequency-domain problem, it requires an FFT to find the time-domain solution.

6.1.2 Theoretical analysis—eigenfunctions

The eigenfunctions solve the problem

$$-\omega^2 \rho \tilde{u} - \partial_x(\mu \partial_x \tilde{u}) = 0. \tag{31}$$

Now, the problem can be represented most straightforwardly using operator notation, ie

$$-\omega^2 \mathcal{P} \left|u\right\rangle + \mathcal{H} \left|u\right\rangle = 0, \tag{32}$$

where we have dropped tildes for ease.

The inner product

We define the inner product as

$$\left\langle u\right| \left|v\right\rangle = \int_0^1 \overline{u}(x) v(x)\,\mathrm{d}x. \tag{33}$$

The operator $\mathcal{P}$

We have

$$\left\langle u\right| \mathcal{P} \left|v\right\rangle = \int_0^1 \overline{u}(x) \rho(x) v(x)\,\mathrm{d}x. \tag{34}$$

As the density is real, it immediately follows that $\mathcal{P}$ is self-adjoint, ie $\mathcal{P}* = \mathcal{P}$.

The operator $\mathcal{H}$

We seek to prove that $\mathcal{H}$ is self-adjoint. We have

$$\begin{aligned}
\left\langle u\right| \mathcal{H} \left|v\right\rangle &= \int_0^1 \overline{u} \partial_x(\mu \partial_x v)\,\mathrm{d}x, \\
&= \mu \overline{u} \partial_x v|_0^1 - \int_0^1 \mu \partial_x \overline{u} \partial_x v\,\mathrm{d}x, \\
&= \mu \overline{u} \partial_x v|_0^1 - \mu \partial_x \overline{u} v|_0^1 + \int_0^1 \partial_x(\mu \partial_x \overline{u}) v\,\mathrm{d}x, \\
&= \mu \overline{u} \partial_x v|_0^1 - \mu \partial_x \overline{u} v|_0^1 + \overline{\left\langle v\right| \mathcal{H} \left|u\right\rangle}.
\end{aligned} \tag{35}$$

For functions satisfying the boundary conditions we specified, ie $u(0) = 0$, and $[\mu u' + u]_1 = 0$ we have for the first two terms

$$\mu(1)\overline{u}(1)\partial_x v|_1 - \mu \partial_x \overline{u}|_1 v(1) = -\mu(1)\overline{u}(1)v(1) + \mu(1)\overline{u}(1)v(1) = 0. \tag{36}$$

Consequently, it follows that $\left\langle u\right| \mathcal{H} \left|v\right\rangle = \overline{\left\langle v\right| \mathcal{H} \left|u\right\rangle}$, ie $\mathcal{H}$ is self-adjoint.

WHAT ABOUT IF THERE ARE DISCONTINUITIES??

Orthogonality

Consider two eigenfunctions

$$-\omega_1^2 \mathcal{P} \left|u_1\right\rangle + \mathcal{H} \left|u_1\right\rangle = 0, \tag{37}$$

$$-\omega_2^2 \mathcal{P} \left|u_2\right\rangle + \mathcal{H} \left|u_2\right\rangle = 0. \tag{38}$$

If we take $\langle u_2|$ with the first equation and $\langle u_1|$ with the second then

$$-\omega_1^2 \left\langle u_2\right| \mathcal{P} \left|u_1\right\rangle + \left\langle u_2\right| \mathcal{H} \left|u_1\right\rangle = 0, \tag{39}$$

$$-\omega_2^2 \left\langle u_1\right| \mathcal{P} \left|u_2\right\rangle + \left\langle u_1\right| \mathcal{H} \left|u_2\right\rangle = 0. \tag{40}$$

Taking the complex conjugate of the second equation, subtracting from the first, and using the self-adjointness of $\mathcal{P}$ and $\mathcal{H}$, it follows that

$$(\overline{\omega_2^2} - \omega_1^2) \left\langle u_2\right| \mathcal{P} \left|u_1\right\rangle = 0. \tag{41}$$

There are two conclusions we can draw:

1. If $\omega_1 \neq \omega_2$, then $\langle u_2| \mathcal{P} |u_1\rangle = 0$, ie eigenfunctions with different eigenfrequencies are orthogonal with respect to the $\mathcal{P}$ inner product.
2. If $|u_1\rangle = |u_2\rangle$ then the inner product is non-zero, and it follows that the eigenfrequencies are real.

6.2 Numerical analysis

6.2.1 The weak form

To obtain the weak form we require a test function $|v\rangle$. If we multiply the strong form (in the frequency domain) by $\overline{v}$, integrate and apply boundary conditions we have

$$\begin{aligned}
-\bar{v}\rho\omega^2 u - \bar{v}\partial_x(\mu\partial_x u) &= \bar{v} f(x,t), \\
-\omega^2 \int_0^1 \rho\bar{v}u \,\mathrm{d}x - \int_0^1 \bar{v}\partial_x(\mu\partial_x u)\,\mathrm{d}x &= \int_0^1 \bar{v} f \,\mathrm{d}x, \\
-\omega^2 \int_0^1 \rho\bar{v}u \,\mathrm{d}x - [\mu\bar{v}\partial_x u]_0^1 + \int_0^1 \mu\partial_x\bar{v}\partial_x u \,\mathrm{d}x &= \int_0^1 \bar{v} f \,\mathrm{d}x, \\
-\omega^2 \int_0^1 \rho\bar{v}u \,\mathrm{d}x + \int_0^1 \mu\partial_x\bar{v}\partial_x u \,\mathrm{d}x + \bar{v}(1)u(1) + \mu(0)\bar{v}(0)\partial_x u(0) &= \int_0^1 \bar{v} f \,\mathrm{d}x,
\end{aligned} \tag{42}$$

using the boundary condition at 1. To impose the boundary condition at $x = 0$ we simply use basis functions that satisfy this condition. In the context of a spectral element mesh this means not including the Lagrange polynomial at zero in the expansion.

6.2.2 The Galerkin principle

If a complete basis $\{|n\rangle\}$ exists, then the wavefield can be expanded as $|u\rangle = \sum_n u_n |n\rangle$. The Galerkin principle states that we can use the basis as our test functions. Consequently

$$\begin{aligned}
M_{nn'} &= \int_0^1 \rho\bar{u}_n u_{n'} \,\mathrm{d}x, \\
H_{nn'} &= \int_0^1 \mu\partial_x\bar{u}_n\partial_x u_{n'} \,\mathrm{d}x + \mu(1)\bar{u}_n(1)\bar{u}_{n'}(1).
\end{aligned} \tag{43}$$

The forced matrix problem is

$$[-\omega^2 M + H]\mathbf{u} = \mathbf{f}. \tag{44}$$

Finally the generalised eigenvalue problem is given by

$$H\mathbf{u} = \omega^2 M\mathbf{u}. \tag{45}$$

6.3 A first example

6.3.1 Analytical approach

As a first example, we consider the problem with a constant $\mu(x) = \mu_0$ and $\rho = 1$. The eigenproblem becomes

$$-\omega^2 u - \mu_0 \partial_x^2 u = 0, \tag{46}$$

subject to $u(0) = 0$, $\mu_0 u'(1) + u(1) = 0$. If we consider the test function $e^{i\lambda x}$ then we have

$$-\omega^2 e^{i\lambda x} + \mu_0 \lambda^2 e^{i\lambda x} = 0. \tag{47}$$

Consequently $\omega^2 = \mu_0 \lambda^2$. To satisfy the boundary condition at $x = 0$ we have $u_n = c_n \sin(\lambda_n x)$. To satisfy the second boundary condition we require

$$\mu_0 \lambda_n \cos(\lambda_n) + \sin(\lambda_n) = 0. \tag{48}$$

This gives

$$\tan(\lambda_n) + \mu_0 \lambda_n = 0. \tag{49}$$

This needs to be solved numerically, using, for instance, the Newton method. For this purpose it is easier to use

$$f(x) = \cot(x) + \frac{1}{\mu_0 x}. \tag{50}$$

Although not obvious, for $n \gg 1$ the solutions become $\lambda_n \to (2n+1)\pi/2$.

6.3.2 Numerical solution

The numerical solution is achieved using a spectral element radial mesh. The radial functions are Lagrange polynomials defined on a GLL grid. For more details see previous set of notes. For the generalised eigenvalue problem we use Spectra, which wraps around Eigen. The result of the calculation for the lowest frequency eigenfunction is shown in Fig. 6. This was performed using 50/150?? elements with 8 GLL points within each element. The pointwise error in the eigenfunction is on the order of 10^{-9}. The eigenfrequency error is on the order of 10^{-9} as well.

6.3.3 Perturbations

If we consider perturbing $\mu_0 \to \mu_1 = 1.1$ we wish to use the eigenfunctions from the first calculation as the basis functions in the perturbed problem. The issue is, however, that the boundary condition at $x = 1$ is not satisfied. Consequently the eigenfunction will be wrong. To get around this one can augment the basis with an arbitrary function. The function chosen was $\sin(\pi x)$ which satisfies the BC at $x = 0$. The result of this calculation is shown in Fig. 7. It can be seen that the spectral element method again is fairly accurate in obtaining the eigenfunction. The expansion using the eigenfunctions from the unperturbed problem clearly does not work particularly well. However, the augmented basis does an excellent job in finding the eigenfunction. Indeed, the augmented basis is correct to 10^{-12} in finding the eigenfrequency, whilst the original basis is off by 10^{-3}.

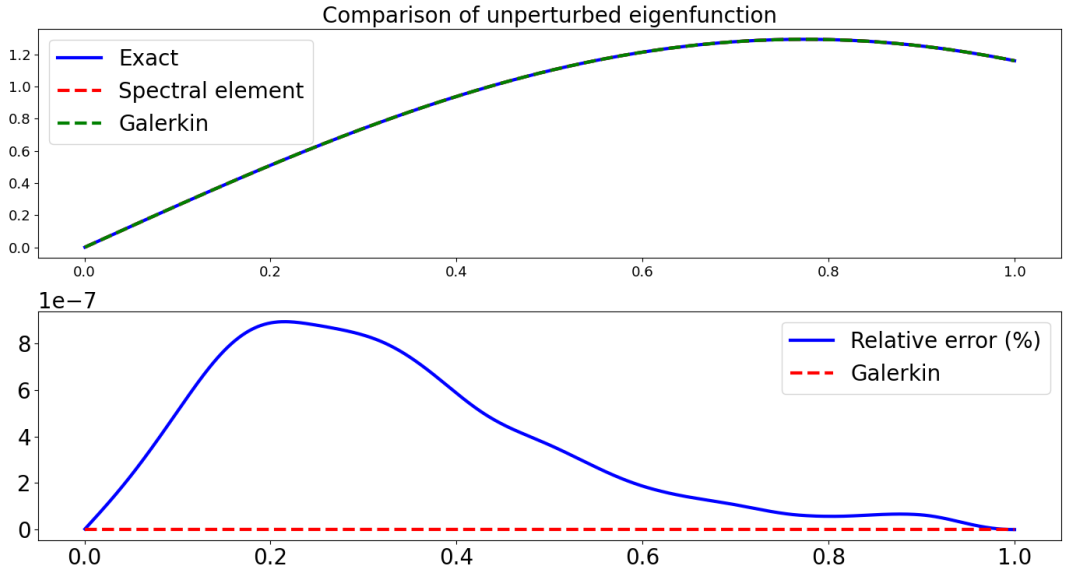

Figure 6: The lowest order eigenfunction for $\mu_0 = 1$. The figure compares the solution obtained from the spectral element Galerkin expansion with the exact result.

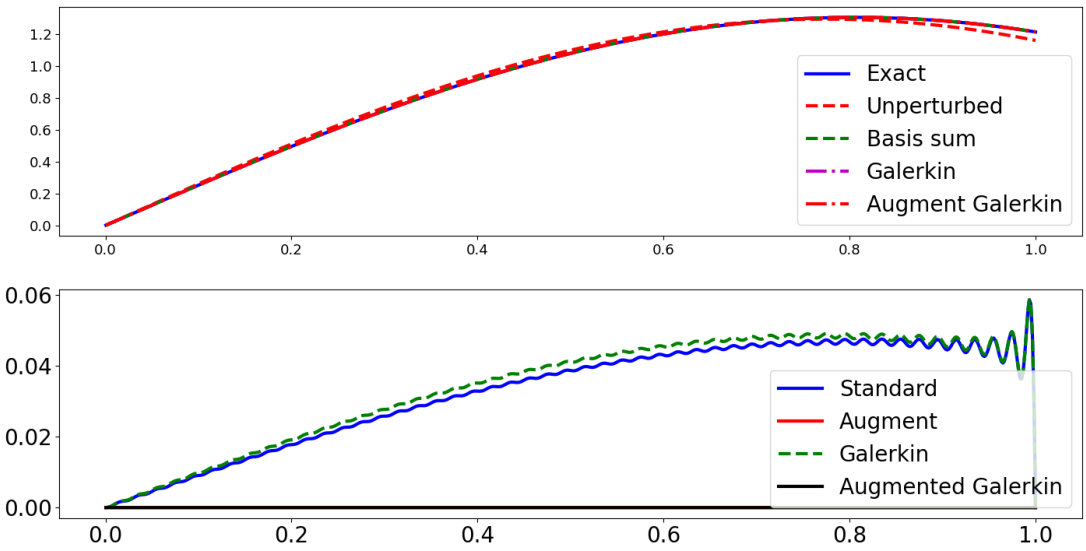

Figure 7: Comparison of different methods for the perturbed problem $\mu_1 = 1.3$.

6.4 A second example

6.4.1 Defining the problem

The second example we will consider is that of two connected homogeneous segments. In particular the domain is decomposed into two sections meeting at $0 < x_m < 1$. The elastic modulus is given by

$$\mu(x) = \begin{cases} \mu_0, 0 < x < x_m, \\ \mu_1, x_m < x < 1. \end{cases} \tag{51}$$

To choose a comparable continuity condition as in the normal mode problem we require that $[u]_-^+ = 0$ and $[\mu u' + u]_-^+ = 0$. Using the first continuity condition this implies $[\mu u']_-^+ = 0$. For the purposes of this problem we choose $\mu_0 = 1$, $\mu_1 = 2$ and $x_m = 0.5$.

6.4.2 Analytical treatment

It is not obvious how one would go about an anlytical treatment of this problem.

6.4.3 Testing code against the exact solution

To test the code we firstly run in the case that $\mu_0 = \mu_1 = 1.0$. In this case, with $N_Q = 6$ and a max element size of 0.05 the eigenfrequency of the gravest mode is accurate to 10^{-11}.

6.4.4 Normal mode basis

We wish to use the normal modes which are obtained from the spectral element solution as the basis functions with a perturbation to the parameters of the model. This means that we need to take in the eigenvectors found by the first part.

To do:

1. Input mu
2. Check continuity condition on boundaries, don't think it is fully implemented right now. In particular, it should be relatively simple, ensuring continuity of derivative, this must be rewriteable
3. Need to think about how to indicate to program where discontinuities are, maybe storing layer indices is a good way?
4. Only want to find the sum term at the discontinuities
5. Needs to be done for the spectral element solver first before putting into new Galerkin solver
6. Input new model
7. Check that same discontinuities
8. Generate matrices
   (a) Need to have derivatives
   (b) Matrices same size as number of basis functions
   (c) Need to have possibility of augmentation

9. Find eigenvectors

10. DONE UP TO HERE NOW, I BELIEVE THAT I HAVE IT CORRECT, BUT NOT 100% SURE. NOW, THE ISSUE IS THAT THE EIGENVECTORS ARE COMPLEX. HOWEVER, EIGEN DOES NOT HAVE THE CAPACITY FOR GENERALISED EIGEN PROBLEMS WITH COMPLEX INPUT MATRICES. CONSEQUENTLY IT PROBABLY MAKES SENSE TO SHIFT DIRECTLY TO SOLVING IN THE FREQUENCY DOMAIN.

11. TO DO:

    (a) ADD AUGMENTATION. THIS SHOULD BE RELATIVELY STRAIGHTFORWARD AS JUST NEED TO GET THE VALUE (AND DERIVATIVES OFC) OF THE AUGMENTED FUNCTIONS INTO THE VAL-EF VECTOR.

12. Solve problem in frequency domain

$$
\begin{aligned}
-\bar{v}\rho\omega^2 u - \bar{v}\partial_x(\mu\partial_x u) &= \bar{v}f(x,t),\\
-\omega^2\int_0^1 \rho\bar{v}u\,\mathrm{d}x - \int_0^1 \bar{v}\partial_x(\mu\partial_x u)\,\mathrm{d}x &= \int_0^1 \bar{v}f\,\mathrm{d}x,\\
-\omega^2\int_0^1 \rho\bar{v}u\,\mathrm{d}x - \sum_e [\mu\bar{v}\partial_x u]_{r_{el}}^{r_{eu}} + \int_0^1 \mu\partial_x\bar{v}\partial_x u\,\mathrm{d}x &= \int_0^1 \bar{v}f\,\mathrm{d}x,\\
-\omega^2\int_0^1 \rho\bar{v}u\,\mathrm{d}x + \int_0^1 \mu\partial_x\bar{v}\partial_x u\,\mathrm{d}x - \mu(1)\bar{v}(1)(\partial_x u)(1)+&\\
+\mu(0)\bar{v}(0)(\partial_x u)(0) - \sum_e [(\mu\bar{v}\partial_x u)(x_e)]_-^+ &= \int_0^1 \bar{v}f\,\mathrm{d}x,\\
-\omega^2\int_0^1 \rho\bar{v}u\,\mathrm{d}x + \int_0^1 \mu\partial_x\bar{v}\partial_x u\,\mathrm{d}x + \bar{v}(1)u(1) + \mu(0)\bar{v}(0)\partial_x u(0) &= \int_0^1 \bar{v}f\,\mathrm{d}x,
\end{aligned}
\tag{52}
$$

The matrices are therefore given by

$$
\begin{aligned}
M_{nn'} &= \int_0^1 \rho\bar{u}_n u_{n'}\,\mathrm{d}x,\\
H_{nn'} &= \int_0^1 \mu\partial_x\bar{u}_n\partial_x u_{n'}\,\mathrm{d}x + \mu(1)\bar{u}_n(1)\bar{u}_{n'}(1) - \sum_e [(\mu\bar{u}_n\partial_x u_{n'})(x_e)]_-^+.
\end{aligned}
\tag{53}
$$

The issue is we have never used the continuity and traction boundary conditions at the internal boundaries. If we are using the eigenfunctions from the unperturbed problem then this is straightforward to implement... the sum over all the discontinuities is zero as $\bar{v}$ is continuous etc. However, if we augment in each layer how do we implement this?

6.4.5 Full analytical benchmark

As a first example, we consider the problem with a constant $\mu(x) = \mu_0$ and $\rho = 1$. The eigenproblem becomes

$$
-\omega^2\rho_0 u - \mu_0\partial_x^2 u = 0, \tag{54}
$$

subject to $u(0) = 0$, $\mu_0 u'(1) + u(1) = 0$. If we consider the test function $e^{i\lambda x}$ then we have

$$-\omega^2 \rho_0 e^{i\lambda x} + \mu_0 \lambda^2 e^{i\lambda x} = 0. \tag{55}$$

Consequently $\omega^2 = (\mu_0/\rho_0)\lambda^2$. To satisfy the boundary condition at $x = 0$ we have $u_n = c_n \sin(\lambda_n x)$. To satisfy the second boundary condition we require

$$\mu_0 \lambda_n \cos(\lambda_n) + \sin(\lambda_n) = 0. \tag{56}$$

This gives

$$\tan(\lambda_n) + \mu_0 \lambda_n = 0. \tag{57}$$

This needs to be solved numerically, using, for instance, the Newton method. For this purpose it is easier to use

$$f(x) = \cot(x) + \frac{1}{\mu_0 x}. \tag{58}$$

Although not obvious, for $n \gg 1$ the solutions become $\lambda_n \to (2n+1)\pi/2$.

7 The Toroidal problem

7.1 The weak form

The 1D toroidal problem can be obtained using either the vector spherical harmonics or generalised spherical harmonics in the full problem. This is detailed elsewhere, e.g. [4]. The toroidal problem is given in the strong form by

$$\dot{W} = r^{-1} W + L^{-1} T, \tag{59}$$

$$\dot{T} = [-\omega^2 \rho + (k^2 - 2) N r^{-2}] W - 3 r^{-1} T, \tag{60}$$

where $T = \mu(\dot{W} - r^{-1} W)$ is the traction. The boundary condition on solid-solid discontinuities is $[W]_-^+ = 0$ and $[T]_-^+$ whilst on fluid-solid boundaries tangential slip is allowed but the traction must vanish. The weak form can be derived straightforwardly from this and is given by

$$\omega^2 \int_\Omega \rho W \tilde{W} r^2 \, \mathrm{d}r = \int_\Omega L(r\partial_r W - W)(r\partial_r \tilde{W} - \tilde{W}) \, \mathrm{d}r + \int_\Omega N(k^2 - 2) W \tilde{W} \, \mathrm{d}r. \tag{61}$$

7.2 Homogeneous sphere benchmark

To solve this we employ the standard spectral element method—following from Kemper et al. The solution for the eigenfrequencies is found using Spectra with the generalised eigen shift solver. Benchmarking is done with the eigenfrequencies of the homogeneous sphere. The analytical condition is given by

$$(l-1) j_l(x) - x j_{l+1}(x) = 0, \tag{62}$$

where $x = \omega a/\beta$ and $j_l(x)$ is the spherical Bessel function of the first kind of degree l. To solve for the eigenfrequencies we use the bisection method as Newton's method requires a more accurate input of the guess due to the oscillation of the Bessel functions.

According to Kemper et al. the convergence is given by h^{2p} where p is the polynomial order in the spectral element method. Benchmarks for $l = 2$ and $l = 5$ are presented in Fig. 8 and Fig. 9. The lines of best fit were found for the log-log plot using polyfit1d in matplotlib. The convergence

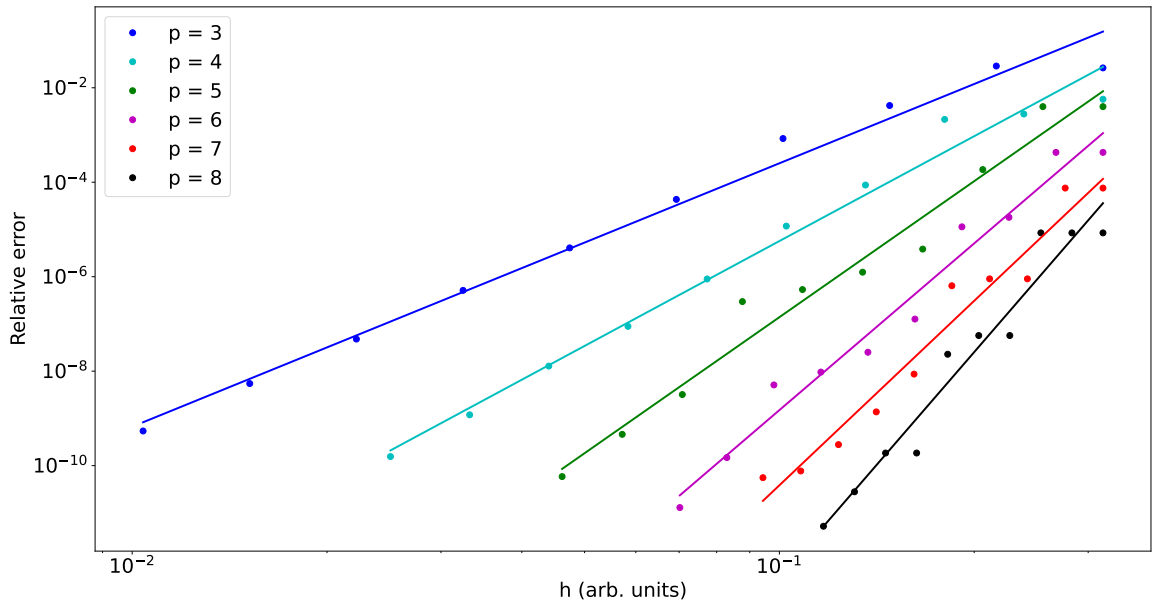

Figure 8: Relative error of eigenvalues from the spectral element method compared to the analytical solution for toroidal modes. The error is calculated for $l = 2$ and I show the mean relative error of the first 10 modes as a function of max element size h.

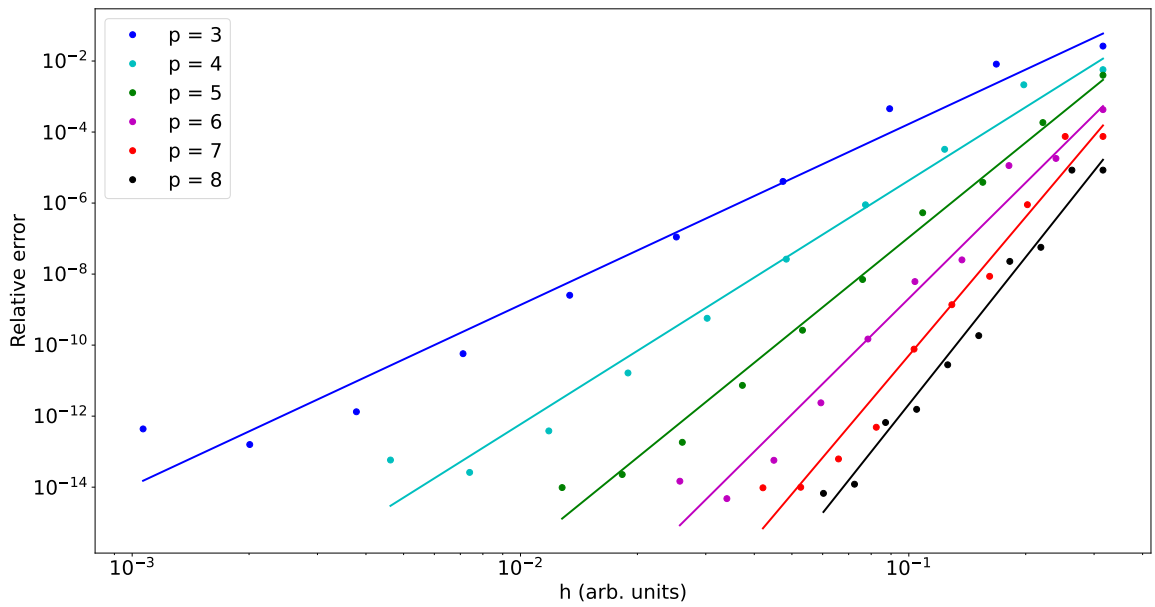

Figure 9: Relative error of eigenvalues from the spectral element method compared to the analytical solution for toroidal modes. The error is calculated for $l = 5$ and I show the mean relative error of the first 10 modes as a function of max element size h.

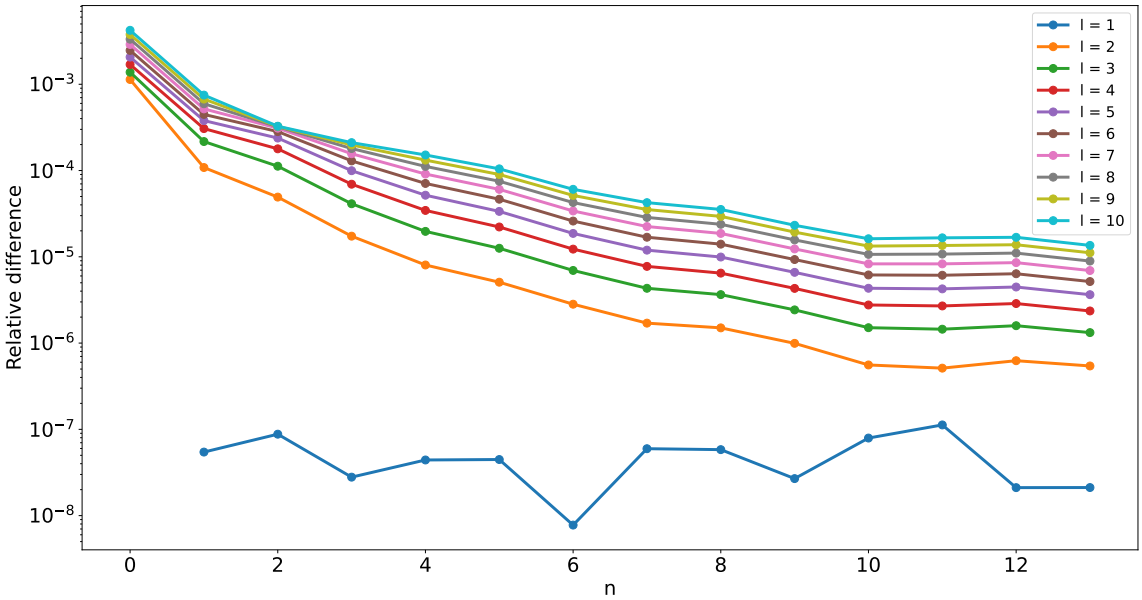

Figure 10: Relative difference between eigenvalues generated from the spectral element method compared to the result from MINEOS for toroidal modes. The calculation is done with a shift in the generalised eigensolver of 0.01, the polynomial order of the spectral element method was 7. The x-axis indicates the overtone number whilst the y-axis is the relative difference between the two methods.

satisfies, approximately, that expected by the h^{2p} relation. For instance for $l = 5$ the exponents of the line of best fit are $5.1, 6.7, 8.9, 10.8, 12.9, 13.8$ for $p = 3, 4, 5, 6, 7, 8$ respectively. This convergence isn't quite as rapid as one may expect but it lies "within reasonable bounds". It should be noted that for smaller elements than shown (for each max polynomial order respectively) the error does not continue decreasing due to numerical errors. This can be seen in both examples for the smallest values of h when $p = 3$.

7.3 MINEOS comparison

We compare the eigenfrequencies found using our program and a nonattenuating PREM model for the toroidal modes of the mantle. This is compared in Fig. 10.

It is interesting to observe the relative difference in this calculation. The difference decreases considerably with increasing n. It isn't necessarily obvious why this would be the case. One potential explanation is the difference in the model associated with the way it is read in. It is striking that for $l = 1$ the difference between the results is on the order of 10^{-7} whilst it is considerably larger for $l > 1$. The reason for this is unclear.

7.4 The traction problem

To illustrate the issue in the traction problem we take the eigenfunctions from PREM (or a version of PREM) and compute the associated traction with a model of PREM perturbed only in the outer

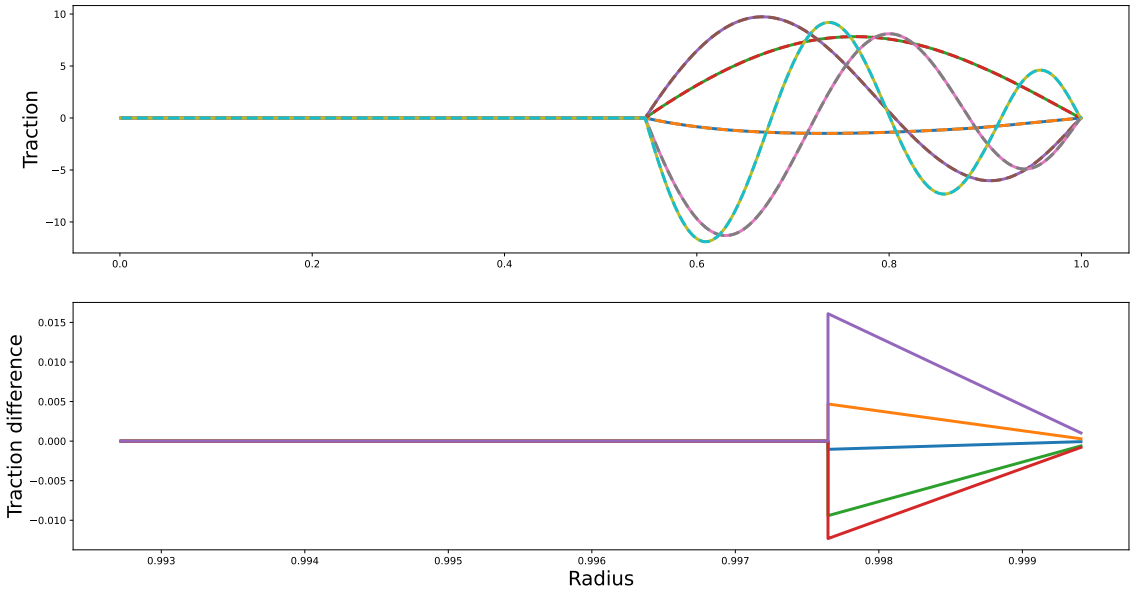

Figure 11: Traction from first 5 toroidal modes ($l = 2$) of PREM in a PREM model perturbted in the outermost layer.

layer. In Fig. 11 the result of this can be seen. The traction is the same for both models except at the outermost boundary. It is here that the model perturbation occurs. Clearly the traction is not continuous for any of the eigenmodes.

7.5 Augmentation

The key issue identified previously is that the normal modes for a particular one-dimensional Earth model are complete in that model. However, they cannot satisfy the boundary conditions. It is reasonable to suppose that we must add a degree of freedom for each component at each boundary and the free surface. To obtain an augmentation basis that is, in some way, natural, we consider the problem. In particular, we have

$$\omega^2 M x = K x, \tag{63}$$

with our boundary conditions being continuity of traction across the boundaries. Perturbations to model parameters induce discontinuity of traction. A basis function that is, in some ways natural, has discontinuity of traction across a boundary. This can be achieved by finding the solution to $Kx = b$, where b is a vector that is zero except at the index corresponding to the boundary under consideration.

We have to be slightly careful. We don't want the augmented functions generated with a traction variation across the CMB or the free surface. The other functions are shown in Fig. 12.

7.6 The results with perturbed PREM

I have constructed a perturbed PREM model with a 2% increase in the shear velocity in the second layer in the mantle as defined in PREM. I have found the normal modes using the Galerkin expansion

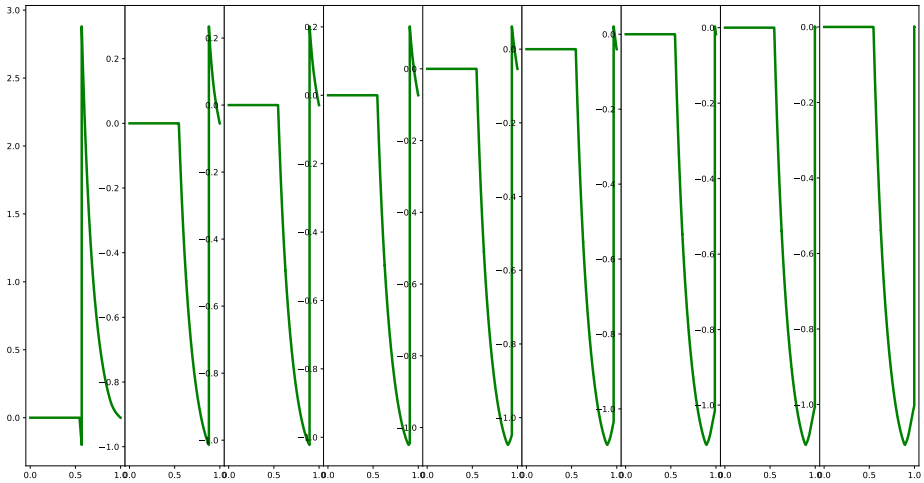

Figure 12: All toroidal augmentation functions

with and without augmenting. The results for the two lowest frequency normal modes are shown in Fig. 13.

The error in the value of the eigenfunctions is on the order of 10^{-5} without augmentation and around an order of magnitude better with augmentation. The error in the eigenfrequency improves by around 2-3 orders of magnitude, from 10^{-7} to $\sim 10^{-10}$.

It should be noted that even without augmentation the eigenfrequencies are pretty close.

Finally I have found the traction in the new model for the eigenfunctions. In Fig. 14 I plot the difference between the traction for the exact modes and the traction calculated using the normal modes from the unperturbed Earth. It clearly shows that the traction is non-continuous without augmentation. Somewhat suprisingly, even with augmentation, the traction is still not continuous, although the difference is much reduced. See the caption for further details.

The big question that arises is how do we ensure that the traction vanishes?

8 The forced problem

8.1 Introduction

To analyse the magnitude of the effect of this problem we want to examine the spectra of the forced problem. In other words if we have a particular Earthquake, what is the ground response? To do this I will take the opportunity to do a full analysis of the equations of motion in terms of the weak form and generalised spherical harmonics. Although perhaps excessive this can be reused in later work and in the thesis...

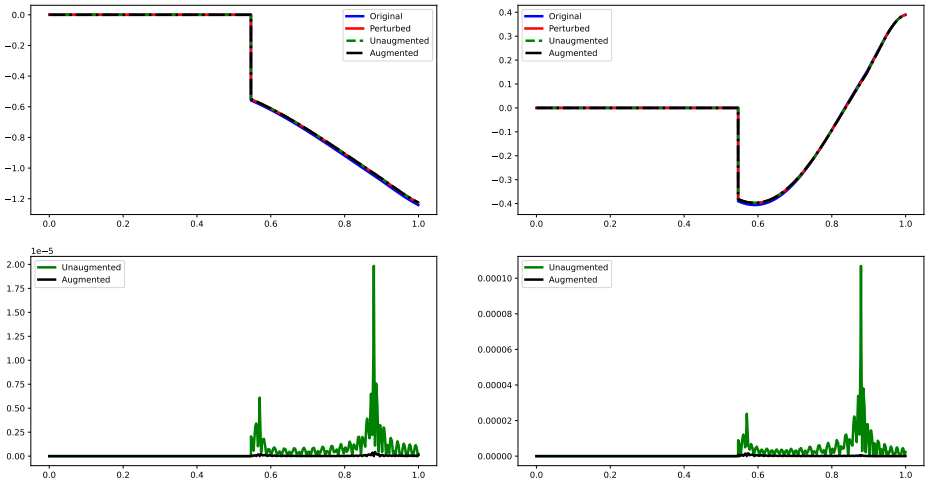

Figure 13: Lowest eigenfrequencies with and without augmentation. Calculations done using 40 normal modes as the basis functions along with the augmentation functions.

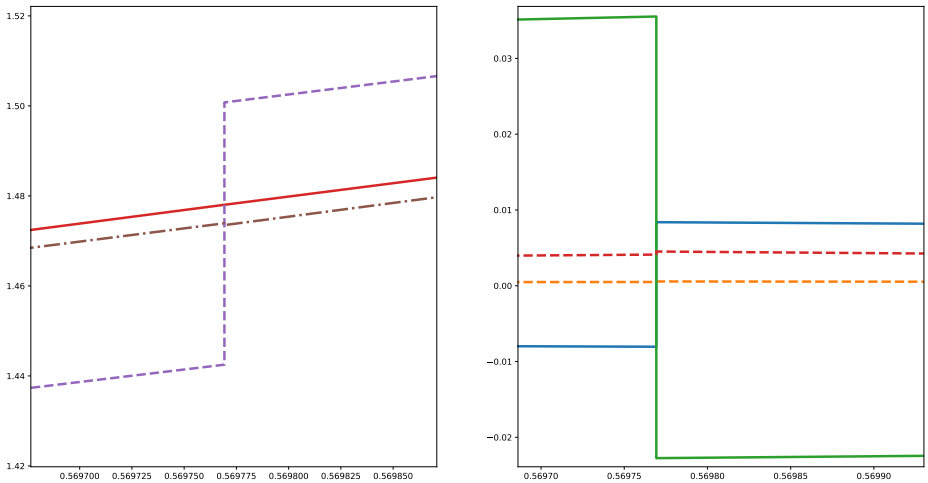

Figure 14: Difference between traction in the perturbed model with the unaugmented and the augmented Galerkin expansions. The left hand figure indicates the traction, with the solid line being for exact, the dashed purple line for the unaugmented solution and the dashed dot brown line being for augmented. In the right hand figure the difference between the traction compared to the exact is shown. The discontinuity is small for the augmented (dashed) but nonzero.

8.2 The equation of motion

8.2.1 Strong form

Our starting point is the equation of motion, as in equation (4.149) of Dahlen and Tromp [4], with the notation of Crawford et al. [3]

$$\rho\partial_t^2\mathbf{u} - \boldsymbol{\nabla}\cdot\mathbf{T} + \boldsymbol{\nabla}(\rho\mathbf{u}\cdot\boldsymbol{\nabla}\Phi) + \rho\boldsymbol{\nabla}\phi + \delta\rho\boldsymbol{\nabla}\Phi = \mathbf{f}, \tag{64}$$

where:

- ρ—density,
- $\mathbf{u}$—particle displacement,
- $\mathbf{T}$—stress tensor,
- Φ—potential in unperturbed body,
- ϕ—perturbation to the potential associated with displacement, given by

$$\nabla^2\phi = -4\pi G\boldsymbol{\nabla}\cdot(\rho\mathbf{u}), \tag{65}$$

- $\delta\rho = -\boldsymbol{\nabla}\cdot(\rho\mathbf{u})$—change in density associated with displacement,
- $\mathbf{f} = -\boldsymbol{\nabla}\cdot\boldsymbol{\Gamma} = -\boldsymbol{\nabla}\cdot[\mathbf{M}\delta(\mathbf{x}-\mathbf{x}_s)H(t)]$—forcing term for an Earthquake located at $\mathbf{x}_s$, occurring at $t_s = 0$, with moment tensor $\mathbf{M}$.

The boundary conditions are

$$\hat{\mathbf{n}}\cdot\mathbf{T} = \hat{\mathbf{n}}\cdot\boldsymbol{\Gamma}, \quad x\in\partial\mathcal{M} \tag{66}$$

$$[\hat{\mathbf{n}}\cdot\mathbf{T}]_-^+ = [\hat{\mathbf{n}}\cdot\boldsymbol{\Gamma}]_-^+, \quad x\in\Sigma_{SS} \tag{67}$$

$$\hat{\mathbf{n}}\cdot\mathbf{T}^+ = \rho^-\left[\mathbf{u}^+\cdot\nabla\Phi+\phi\right]\hat{\mathbf{n}} + \hat{\mathbf{n}}\cdot\boldsymbol{\Gamma}^+, \quad x\in\Sigma_{FS} \tag{68}$$

$$\hat{\mathbf{n}}\cdot\mathbf{T}^- = \rho^+\left[\mathbf{u}^-\cdot\nabla\Phi+\phi\right]\hat{\mathbf{n}} + \hat{\mathbf{n}}\cdot\boldsymbol{\Gamma}^-, \quad x\in\Sigma_{SF} \tag{69}$$

$$[\mathbf{u}]_-^+ = 0, \quad x\in\Sigma_{SS} \tag{70}$$

$$[\phi]_-^+ = 0, \quad x\in\Sigma \tag{71}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi\right]_-^+ - \rho^-\hat{\mathbf{n}}\cdot\mathbf{u}^- = 0, \quad x\in\partial M \tag{72}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi + \rho\hat{\mathbf{n}}\cdot\mathbf{u}\right]_-^+ = 0, \quad x\in\Sigma_{SS} \tag{73}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi\right]_-^+ + [\rho]_-^+\,\hat{\mathbf{n}}\cdot\mathbf{u}^+ = 0, \quad x\in\Sigma_{FS} \tag{74}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi\right]_-^+ + [\rho]_-^+\,\hat{\mathbf{n}}\cdot\mathbf{u}^- = 0, \quad x\in\Sigma_{SF} \tag{75}$$

8.2.2 Weak form

We wish to find the weak form of this equation. We choose the test function $\mathbf{w}$ and pre-multiply by its complex conjugate, before integrating over the body.

First term

We have

$$-\int_{\mathcal{M}}\mathbf{w}^\dagger\omega^2\rho\mathbf{u}\,\mathrm{d}V = -\omega^2\int_{\mathcal{M}}\rho\mathbf{w}^\dagger\mathbf{u}\,\mathrm{d}V. \tag{76}$$

Second term

$$
\begin{aligned}
-\int_{\mathcal{M}} \mathbf{w}^\dagger \cdot [\boldsymbol{\nabla} \cdot \mathbf{T}]\,\mathrm{d}V &= -\int_{\mathcal{M}} \boldsymbol{\nabla} \cdot [\mathbf{w}^\dagger \cdot \mathbf{T}]\,\mathrm{d}V + \int_{\mathcal{M}} \boldsymbol{\nabla}\mathbf{w}^\dagger : \mathbf{T}^T\,\mathrm{d}V, \\
&= -\int_{\partial\mathcal{M}} \mathbf{w}^\dagger \cdot \mathbf{T} \cdot \hat{\mathbf{n}}\,\mathrm{d}S + \int_{\mathcal{M}} \boldsymbol{\nabla}\mathbf{w}^\dagger : \mathbf{T}^T\,\mathrm{d}V. \qquad (77)
\end{aligned}
$$

Third term

$$
\begin{aligned}
\int_{\mathcal{M}} \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}(\rho\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V &= \int_{\mathcal{M}} \boldsymbol{\nabla} \cdot (\mathbf{w}^\dagger \rho\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V - \int_{\mathcal{M}} \rho(\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)(\boldsymbol{\nabla} \cdot \mathbf{w}^\dagger)\,\mathrm{d}V, \\
&= \int_{\partial\mathcal{M}} (\mathbf{w}^\dagger \cdot \hat{\mathbf{n}})(\rho\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}S - \int_{\mathcal{M}} \rho(\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)(\boldsymbol{\nabla} \cdot \mathbf{w}^\dagger)\,\mathrm{d}V \qquad (78)
\end{aligned}
$$

Force term

Applying Gauss' law to the force term we get

$$
\int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^\dagger\,\mathrm{d}V. \qquad (79)
$$

Overall

I am not sure about the convention used in the work for the divergence because it feels like it needs to be with the first coefficient not the second based on how things are defined. All we need to worry about is that the continuity condition on all boundaries is $[\hat{\mathbf{n}} \cdot (\mathbf{T} - \boldsymbol{\Gamma})]_-^+ = 0$, and on free surface the traction and force above the surface are obviously zero. Consequently, we find

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{M}} \rho\mathbf{w}^\dagger\mathbf{u}\,\mathrm{d}V + \int_{\mathcal{M}} \mathbf{T} : \boldsymbol{\nabla}\mathbf{w}^\dagger\,\mathrm{d}V + \int_{\mathcal{M}} \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}(\rho\mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V \\
&+ \int_{\mathcal{M}} \rho\mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\phi\,\mathrm{d}V - \int_{\mathcal{M}} [\mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\Phi][\boldsymbol{\nabla} \cdot (\rho\mathbf{u})]\,\mathrm{d}V = \int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^\dagger\,\mathrm{d}V \qquad (80)
\end{aligned}
$$

8.2.3 Moment tensor point source

A moment tensor point source is given (in the time domain) by

$$
\boldsymbol{\Gamma} = \mathbf{M}\delta(\mathbf{x} - \mathbf{x}_s)H(t), \qquad (81)
$$

where $\mathbf{M}$ is the moment tensor, $\mathbf{x}_s$ is the source location and $H(t)$ is the Heaviside step function, representing an Earthquake occurring at $t = 0$. The force is then given by

$$
\begin{aligned}
\int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^\dagger\,\mathrm{d}V &= H(t) \int_{\mathcal{M}} \delta(\mathbf{x} - \mathbf{x}_s)\mathbf{M} : \boldsymbol{\nabla}\mathbf{w}^\dagger\,\mathrm{d}V, \\
&= H(t)\ \mathbf{M} : \boldsymbol{\nabla}\mathbf{w}^\dagger\big|_{\mathbf{x}=\mathbf{x}_s}. \qquad (82)
\end{aligned}
$$

In the frequency domain the Heaviside step function will be given by

$$\begin{aligned} H(\omega) &= \int_0^\infty e^{-i\omega t}\,\mathrm{d}t, \\ &= -\frac{1}{i\omega} e^{-i\omega t}|_0^\infty, \\ &= -\frac{i}{\omega}, \end{aligned} \tag{83}$$

if $\omega = \omega - i\epsilon$ where $\epsilon > 0$ is a small shift parameter.

8.3 Reduction to 1D

8.3.1 GSH setup

Test function

We now take the specific form of the test function $\mathbf{w} = w^\alpha_{lm} Y^\alpha_{lm} \hat{\mathbf{e}}_\alpha$.

Displacement

The displacement $\mathbf{u}$ is written in terms of generalised spherical harmonics, ie

$$\mathbf{u} = \sum_{lm\alpha} u^\alpha_{lm} Y^\alpha_{lm} \hat{\mathbf{e}}_\alpha. \tag{84}$$

Now, the expansion coefficients must satisfy a few relations. In particular, taking the $\pm$ components,

$$\mathbf{u} = \sum_{lm} u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+. \tag{85}$$

As $\mathbf{u} = \mathbf{u}^*$ we have

$$\begin{aligned} \sum_{lm} u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+ &= \sum_{lm} \overline{u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+}, \\ \sum_{lm} u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+ &= \sum_{lm} \overline{u^-_{lm}} (-1)^{m-1} Y^+_{l,-m} \overline{\hat{\mathbf{e}}_-} + \overline{u^+_{lm}} (-1)^{m+1} Y^-_{l,-m} \overline{\hat{\mathbf{e}}_+}, \\ \sum_{lm} u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+ &= \sum_{lm} (-1)^m \left[\overline{u^-_{lm}} Y^+_{l,-m} \hat{\mathbf{e}}_+ + \overline{u^+_{lm}} Y^-_{l,-m} \hat{\mathbf{e}}_- \right]. \end{aligned} \tag{86}$$

As the sum over m is from $m = -l \to l$ we can take $m \to -m$ and hence

$$\sum_{lm} \left[u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+ \right] = \sum_{lm} (-1)^m \left[\overline{u^-_{l,-m}} Y^+_{l,m} \hat{\mathbf{e}}_+ + \overline{u^+_{l,-m}} Y^-_{l,m} \hat{\mathbf{e}}_- \right]. \tag{87}$$

This is satisfied in general if and only if

$$\overline{u^+_{lm}} = (-1)^m u^-_{l,-m}, \tag{88}$$

$$\overline{u^-_{lm}} = (-1)^m u^+_{l,-m}. \tag{89}$$

Standard decomposition

It is fairly common to use a slight variant of this decomposition. In particular, we can write $u^{\pm}_{lm} = \frac{1}{\sqrt{2}}(V_{lm} \pm iW_{lm})$ and $u^0_{lm} = U_{lm}$, giving

$$\mathbf{u} = \sum_{lm} U_{lm} Y^0_{lm} \hat{\mathbf{e}}_0 + \frac{1}{\sqrt{2}} \sum_{lm} V_{lm} \left[Y^-_{lm} \hat{\mathbf{e}}_- + Y^+_{lm} \hat{\mathbf{e}}_+ \right] + \frac{i}{\sqrt{2}} \sum_{lm} W_{lm} \left[Y^+_{lm} \hat{\mathbf{e}}_+ - Y^-_{lm} \hat{\mathbf{e}}_- \right]. \tag{90}$$

We write

$$\mathbf{w} = \frac{i}{\sqrt{2}} \sum_{lm} W_{lm} \left[Y^+_{lm} \hat{\mathbf{e}}_+ - Y^-_{lm} \hat{\mathbf{e}}_- \right]. \tag{91}$$

It is trivial to show that $\mathbf{w} = \overline{\mathbf{w}}$, as $\overline{W_{lm}} = (-1)^m W_{l,-m}$ and $\overline{\mathbf{Y}^N_{lm}} = (-1)^m \mathbf{Y}^{-N}_{l,-m}$.

The vector $\mathbf{w}$ is the toroidal oscillation component. This decomposition will be used post derivations which employ GSHs.

Stress tensor

The stress tensor is given by

$$\mathbf{T} = \left(\kappa - \frac{2}{3}\mu \right) (\boldsymbol{\nabla} \cdot \mathbf{u}) \mathbf{I} + \mu [\boldsymbol{\nabla} \mathbf{u} + (\boldsymbol{\nabla} \mathbf{u})^T] = \left(\kappa - \frac{2}{3}\mu \right) (\boldsymbol{\nabla} \cdot \mathbf{u}) \mathbf{I} + 2\mu \boldsymbol{\epsilon}, \tag{92}$$

where κ is the bulk modulus and μ is the shear modulus. Using the rules in Dahlen and Tromp [4] we have

$$\boldsymbol{\nabla} \cdot \mathbf{u} = \sum_{lm} [\partial_r u^0_{lm} + 2r^{-1} u^0_{lm} - \Omega^0_l r^{-1} (u^-_{lm} + u^+_{lm})] Y^0_{lm}. \tag{93}$$

The identity tensor is

$$\mathbf{I} = -\hat{\mathbf{e}}_- \otimes \hat{\mathbf{e}}_+ + \hat{\mathbf{e}}_0 \otimes \hat{\mathbf{e}}_0 - \hat{\mathbf{e}}_+ \otimes \hat{\mathbf{e}}_-. \tag{94}$$

The strain is given by

$$\begin{aligned} \boldsymbol{\epsilon} = \sum_{lm} \Bigg\{ & (\partial_r u^0_{lm}) Y^0_{lm} (\hat{\mathbf{e}}_0 \otimes \hat{\mathbf{e}}_0) + \Omega^2_l r^{-1} u^{\pm}_{lm} Y^{\pm 2}_{lm} (\hat{\mathbf{e}}_{\pm} \otimes \hat{\mathbf{e}}_{\pm}) \\ & + \frac{1}{2} \left(\frac{du^{\pm}_{lm}}{dr} - \frac{u^{\pm}_{lm}}{r} + \frac{\Omega^0_l u^0_{lm}}{r} \right) Y^{\pm}_{lm} [(\hat{\mathbf{e}}_0 \otimes \hat{\mathbf{e}}_{\pm}) + (\hat{\mathbf{e}}_{\pm} \otimes \hat{\mathbf{e}}_0)] + \\ & + \left[\frac{1}{2} \Omega^0_l r^{-1} (u^-_{lm} + u^+_{lm}) - r^{-1} u^0_{lm} \right] Y^0_{lm} \left[(\hat{\mathbf{e}}_+ \otimes \hat{\mathbf{e}}_-) + (\hat{\mathbf{e}}_- \otimes \hat{\mathbf{e}}_+) \right] \Bigg\}, \end{aligned} \tag{95}$$

where $\pm$ is shorthand for summing over both the positive and negative components. The components of $\mathbf{T}$ are

$$T^{00} = \left(\kappa - \frac{2}{3}\mu\right)[\partial_r u^0_{lm} + 2r^{-1}u^0_{lm} - \Omega^0_l r^{-1}(u^-_{lm} + u^+_{lm})] + 2\mu\partial_r u^0_{lm}, \tag{96}$$

$$T^{\pm\pm} = 2\mu\Omega_l^2 r^{-1}u^\pm_{lm}, \tag{97}$$

$$T^{\pm 0} = \mu\left(\frac{du^\pm_{lm}}{dr} - \frac{u^\pm_{lm}}{r} + \frac{\Omega^0_l u^0_{lm}}{r}\right), \tag{98}$$

$$T^{0\pm} = \mu\left(\frac{du^\pm_{lm}}{dr} - \frac{u^\pm_{lm}}{r} + \frac{\Omega^0_l u^0_{lm}}{r}\right), \tag{99}$$

$$T^{+-} = 2\mu\left[\frac{1}{2}\Omega^0_l r^{-1}(u^-_{lm} + u^+_{lm}) - r^{-1}u^0_{lm}\right] - \left(\kappa - \frac{2}{3}\mu\right)[\dot{u}^0_{lm} + 2r^{-1}u^0_{lm} - \Omega^0_l r^{-1}(u^-_{lm} + u^+_{lm})], \tag{100}$$

$$T^{-+} = 2\mu\left[\frac{1}{2}\Omega^0_l r^{-1}(u^-_{lm} + u^+_{lm}) - r^{-1}u^0_{lm}\right] - \left(\kappa - \frac{2}{3}\mu\right)[\dot{u}^0_{lm} + 2r^{-1}u^0_{lm} - \Omega^0_l r^{-1}(u^-_{lm} + u^+_{lm})]. \tag{101}$$

Simplifying and rewriting

$$T^{00} = (\kappa + \tfrac{4}{3}\mu)\dot{u}^0_{lm} + (\kappa - \tfrac{2}{3}\mu)r^{-1}[2u^0_{lm} - \Omega^0_l(u^-_{lm} + u^+_{lm})], \tag{102}$$

$$T^{\pm\pm} = 2\mu\Omega_l^2 r^{-1}u^\pm_{lm}, \tag{103}$$

$$T^{\pm 0} = T^{0\pm} = \mu\left[\dot{u}^\pm_{lm} - r^{-1}(u^\pm_{lm} - \Omega^0_l u^0_{lm})\right], \tag{104}$$

$$T^{\pm\mp} = -(\kappa - \tfrac{2}{3}\mu)\dot{u}^0_{lm} - (\kappa + \tfrac{1}{3}\mu)r^{-1}\left[2u^0_{lm} - \Omega^0_l(u^-_{lm} + u^+_{lm})\right], \tag{105}$$

where overdots indicate radial derivatives.

Model parameters

The potential in the background model Φ is given by solving Poisson's equation, with

$$\nabla^2\Phi = 4\pi G\rho. \tag{106}$$

As we are doing 1D calculations, we find $\Phi = \Phi(r)$, consequently

$$\boldsymbol{\nabla}\Phi = g(r)\hat{\mathbf{n}}, \tag{107}$$

where

$$g(r) = \frac{GM(r)}{r^2}, \tag{108}$$

and $M(r)$ is the mass contained within the radius r. For most of the calculations presented below we simply use g and assume that it is solved separately. The gravitational acceleration satisfies a simple 1D differential equation:

$$\begin{aligned} \boldsymbol{\nabla}\cdot(g\hat{\mathbf{n}}) &= 4\pi G\rho, \\ \frac{1}{r^2}\frac{\partial(r^2 g)}{\partial r} &= 4\pi G\rho, \\ \dot{g} + 2r^{-1}g &= 4\pi G\rho. \end{aligned} \tag{109}$$

Moment tensor

The force is given by

$$\mathbf{f} = -\boldsymbol{\nabla} \cdot \boldsymbol{\Gamma}, \tag{110}$$

where

$$\boldsymbol{\Gamma} = \mathbf{M}\delta(\mathbf{x} - \mathbf{x}_s)H(t), \tag{111}$$

where $\mathbf{M}$ is the moment tensor, $\mathbf{x}_s$ is the source location and $H(t)$ is the Heaviside step function. Now, typically the moment tensor is given by

$$\begin{aligned}\mathbf{M} = M_{rr}(\hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + M_{r\theta}(\hat{\mathbf{r}} \otimes \hat{\boldsymbol{\theta}}) + M_{r\varphi}(\hat{\mathbf{r}} \otimes \hat{\boldsymbol{\varphi}}) + M_{\theta r}(\hat{\boldsymbol{\theta}} \otimes \hat{\mathbf{r}}) + M_{\theta\theta}(\hat{\boldsymbol{\theta}} \otimes \hat{\boldsymbol{\theta}}) + M_{\theta\varphi}(\hat{\boldsymbol{\theta}} \otimes \hat{\boldsymbol{\varphi}}) + \\ + M_{\varphi r}(\hat{\boldsymbol{\varphi}} \otimes \hat{\mathbf{r}}) + M_{\varphi\theta}(\hat{\boldsymbol{\varphi}} \otimes \hat{\boldsymbol{\theta}}) + M_{\varphi\varphi}(\hat{\boldsymbol{\varphi}} \otimes \hat{\boldsymbol{\varphi}}).\end{aligned} \tag{112}$$

Now, the canonical basis vectors are

$$\hat{\mathbf{e}}_- = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{\theta}} - i\hat{\boldsymbol{\varphi}}), \tag{113}$$

$$\hat{\mathbf{e}}_0 = \hat{\mathbf{r}}, \tag{114}$$

$$\hat{\mathbf{e}}_+ = -\frac{1}{\sqrt{2}}(\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\varphi}}). \tag{115}$$

Consequently, we can write

$$\hat{\mathbf{r}} = \hat{\mathbf{e}}_0, \tag{116}$$

$$\hat{\boldsymbol{\theta}} = \frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_- - \hat{\mathbf{e}}_+), \tag{117}$$

$$\hat{\boldsymbol{\varphi}} = \frac{i}{\sqrt{2}}(\hat{\mathbf{e}}_- + \hat{\mathbf{e}}_+). \tag{118}$$

In terms of the canonical basis we need to have

$$\mathbf{M} = \sum_{\alpha\beta} M^{\alpha\beta}(\hat{\mathbf{e}}_\alpha \otimes \hat{\mathbf{e}}_\beta). \tag{119}$$

The expansion coefficients are

$$M^{00} = M_{rr}, \tag{120}$$

$$M^{--} = \frac{1}{2}(M_{\theta\theta} - M_{\varphi\varphi} + iM_{\theta\varphi} + iM_{\varphi\theta}), \tag{121}$$

$$M^{++} = \frac{1}{2}(M_{\theta\theta} - M_{\varphi\varphi} - iM_{\theta\varphi} - iM_{\varphi\theta}), \tag{122}$$

$$M^{-0} = \frac{1}{\sqrt{2}}(M_{\theta r} + iM_{\varphi r}), \tag{123}$$

$$M^{+0} = -\frac{1}{\sqrt{2}}(M_{\theta r} - iM_{\varphi r}), \tag{124}$$

$$M^{0-} = \frac{1}{\sqrt{2}}(M_{r\theta} + iM_{r\varphi}), \tag{125}$$

$$M^{0+} = -\frac{1}{\sqrt{2}}(M_{r\theta} - iM_{r\varphi}), \tag{126}$$

$$M^{-+} = \frac{1}{2}(-M_{\theta\theta} - M_{\varphi\varphi} + iM_{\theta\varphi} - iM_{\varphi\theta}), \tag{127}$$

$$M^{+-} = \frac{1}{2}(-M_{\theta\theta} - M_{\varphi\varphi} - iM_{\theta\varphi} + iM_{\varphi\theta}). \tag{128}$$

If the moment tensor is symmetric (as almost always is the case in seismological applications) then $M_{r\theta} = M_{\theta r}$, $M_{r\varphi} = M_{\varphi r}$ and $M_{\theta\varphi} = M_{\varphi\theta}$. Using this and simplifying with $\pm$, we have

$$M^{00} = M_{rr}, \tag{129}$$

$$M^{\pm 0} = M^{0\pm} = \frac{1}{\sqrt{2}}(\mp M_{\theta r} + iM_{\varphi r}), \tag{130}$$

$$M^{\pm\pm} = \frac{1}{2}(M_{\theta\theta} - M_{\varphi\varphi} \mp 2iM_{\theta\varphi}), \tag{131}$$

$$M^{\pm\mp} = -\frac{1}{2}(M_{\theta\theta} + M_{\varphi\varphi}). \tag{132}$$

8.3.2 The 1D reduction

To make it clear in the following derivation we will make summations over upper indices of the generalised spherical harmonic components or over l and m explicit. At times we will not indicate the dependence on r, θ, φ, although if it is not trivial, or aids understanding, this will likely be indicated. We also express the domain of the radial integration by $\mathcal{R}$

Term 1

The first term is

$$\begin{aligned} -\omega^2 \int_{\mathcal{M}} \rho \mathbf{w}^\dagger \mathbf{u} \, \mathrm{d}V &= -\omega^2 \int_{\mathcal{M}} \rho(r) \overline{w^\alpha_{lm}(r) Y^\alpha_{lm}} \hat{\mathbf{e}}^\alpha \cdot \mathbf{u} \, \mathrm{d}V, \\ &= -\omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w^\alpha_{lm}} u^\alpha_{lm} \, \mathrm{d}r. \end{aligned} \tag{133}$$

Term 2

The second term is $\int_{\mathcal{M}} \mathbf{T} : \boldsymbol{\nabla}\mathbf{w}^\dagger \, \mathrm{d}V$. The gradient of test function needs to be rewritten as follows:

$$\begin{aligned} \boldsymbol{\nabla}\mathbf{w}^\dagger &= \overline{\boldsymbol{\nabla}\mathbf{w}}, \\ &= \overline{\boldsymbol{\nabla}[w^\alpha_{lm} Y^\alpha_{lm} \hat{\mathbf{e}}_\alpha]}. \end{aligned} \tag{134}$$

It is easiest to consider the cases of $\alpha = 0$ and $\alpha = \pm$ separately. We write $\mathbf{G} = \boldsymbol{\nabla}\mathbf{w}$, and write out the components of $\mathbf{G}$ based on Dahlen and Tromp [4].

$\boldsymbol{\alpha} = \mathbf{0}$:

$$G^{00} = \dot{w}^0_{lm}, \tag{135}$$

$$G^{\pm 0} = r^{-1}\Omega^0_l w^0_{lm}, \tag{136}$$

$$G^{\pm\mp} = -r^{-1} w^0_{lm}. \tag{137}$$

$\boldsymbol{\alpha} = \pm$:

$$G^{0\pm} = \dot{w}^\pm_{lm}, \tag{138}$$

$$G^{\pm 0} = -r^{-1} w^\pm_{lm}, \tag{139}$$

$$G^{\pm\pm} = \Omega^2_l r^{-1} w^\pm_{lm}, \tag{140}$$

$$G^{\mp\pm} = r^{-1}\Omega^0_l w^\pm_{lm}. \tag{141}$$

The usage of the above notation is designed to reduce the total amount of writing. The way to use the above for a particular test function is simply to read the top or bottom line in the $\pm$ symbols.

Now, we need to find

$$\begin{aligned} \mathbf{T} : \overline{\boldsymbol{\nabla}\mathbf{w}} &= \sum_{\alpha\beta} T^{\alpha\beta}(\hat{\mathbf{e}}_\alpha \otimes \hat{\mathbf{e}}_\beta) \overline{\sum_{\alpha'\beta'} G^{\alpha'\beta'}(\hat{\mathbf{e}}_{\alpha'} \otimes \hat{\mathbf{e}}_{\beta'})}, \\ &= \sum_{\alpha\beta} T^{\alpha\beta} \overline{G^{\alpha\beta}}, \end{aligned} \tag{142}$$

due to the orthogonality of the canonical basis. When we consider the integral and expand in (lm) we find

$$\begin{aligned} \int_{\mathcal{M}} \mathbf{T} : \overline{\boldsymbol{\nabla}\mathbf{w}} &= \sum_{\alpha\beta} \sum_{lm} \sum_{l'm'} \int_{\mathcal{M}} T^{\alpha\beta}_{lm}(r) Y^{\alpha+\beta}_{lm} \overline{G^{\alpha\beta}_{l'm'}(r) Y^{\alpha+\beta}_{l'm'}} \, \mathrm{d}V, \\ &= \sum_{\alpha\beta} \sum_{lm} \int_{\mathcal{R}} r^2 T^{\alpha\beta}_{lm} \overline{G^{\alpha\beta}_{lm}} \, \mathrm{d}V. \end{aligned} \tag{143}$$

As $\mathbf{w} = w^\alpha_{lm} Y^\alpha_{lm} \hat{\mathbf{e}}_\alpha$ we no longer have the sum over lm.

$\boldsymbol{\alpha} = \mathbf{0}$:

$$T_2 = \int_{\mathcal{R}} r^2 \left\{ \overline{\dot{w}^0_{lm}} T^{00}_{lm} + r^{-1} w^0_{lm} \left[\Omega^0_l (T^{+0}_{lm} + T^{-0}_{lm}) - (T^{+-}_{lm} + T^{-+}_{lm}) \right] \right\} \mathrm{d}r. \tag{144}$$

$\boldsymbol{\alpha} = \pm$:

$$T_2 = \int_{\mathcal{R}} r^2 \left\{ \overline{\dot{w}^\pm_{lm}} T^{0\pm}_{lm} + r^{-1} \overline{w^\pm_{lm}} \left[\Omega^2_l T^{\pm\pm} + \Omega^0_l T^{\mp\pm} - T^{\pm 0}_{lm} \right] \right\} \mathrm{d}r. \tag{145}$$

Term 3

We have

$$\int_{\mathcal{M}} \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}(\rho \mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V. \tag{146}$$

The potential Φ is a function of r only, so we have

$$\begin{aligned} \rho \mathbf{u} \cdot \boldsymbol{\nabla}\Phi &= \rho \mathbf{u} \cdot g(r)\hat{\mathbf{r}}, \\ &= \sum_{lm} [\rho(r) g(r) u^0_{lm}(r)] Y^0_{lm}. \end{aligned} \tag{147}$$

Consequently, the gradient is

$$\begin{aligned} \boldsymbol{\nabla}(\rho \mathbf{u} \cdot \boldsymbol{\nabla}\Phi) &= \boldsymbol{\nabla}\left\{\sum_{lm} [\rho(r) g(r) u^0_{lm}(r)] Y^0_{lm}\right\}, \\ &= \sum_{lm} \left\{[\partial_r(\rho g u^0_{lm}) Y^0_{lm} \hat{\mathbf{e}}_0 + \Omega^0_l r^{-1} u^0_{lm} \left[Y^+_{lm}\hat{\mathbf{e}}_+ + Y^-_{lm}\hat{\mathbf{e}}_-\right]]\right\}. \end{aligned} \tag{148}$$

Finally, for the two options, $\alpha = 0$ or $\alpha = \pm$, we have

$\boldsymbol{\alpha = 0}$:

$$\int_{\mathcal{M}} \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}(\rho \mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V = \int_{\mathcal{R}} r^2 \left[\overline{w^0_{lm}} \cdot \frac{d}{dr}(\rho g u^0_{lm})\right] \mathrm{d}r \tag{149}$$

$\boldsymbol{\alpha = \pm}$:

$$\int_{\mathcal{M}} \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}(\rho \mathbf{u} \cdot \boldsymbol{\nabla}\Phi)\,\mathrm{d}V = \Omega^0_l \int_{\mathcal{R}} \rho g r \overline{w^\pm_{lm}} u^0_{lm}\,\mathrm{d}r \tag{150}$$

Term 4

We have

$$\int_{\mathcal{M}} \rho \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\phi\,\mathrm{d}V. \tag{151}$$

We will need an additional equation for the potential ϕ. We simply write

$$\phi = \sum_{lm} P_{lm} Y^0_{lm}, \tag{152}$$

which gives

$$\boldsymbol{\nabla}\phi = \sum_{lm} \left[\dot{P}_{lm} Y^0_{lm} \hat{\mathbf{e}}_0 + \Omega^0_l r^{-1} P_{lm} (Y^+_{lm}\hat{\mathbf{e}}_+ + Y^-_{lm}\hat{\mathbf{e}}_-)\right]. \tag{153}$$

$\boldsymbol{\alpha = 0}$:

$$\int_{\mathcal{M}} \rho \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\phi\,\mathrm{d}V = \int_{\mathcal{R}} r^2 \rho \overline{w^0_{lm}} \dot{P}_{lm}\,\mathrm{d}r. \tag{154}$$

$\boldsymbol{\alpha = \pm}$:

$$\int_{\mathcal{M}} \rho \mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\phi\,\mathrm{d}V = \Omega^0_l \int_{\mathcal{R}} r \rho \overline{w^\pm_{lm}} P_{lm}\,\mathrm{d}r. \tag{155}$$

Term 5

The term is

$$\int_{\mathcal{M}} (\mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\Phi)[\boldsymbol{\nabla} \cdot (\rho \mathbf{u})] \, \mathrm{d}V. \tag{156}$$

As $\boldsymbol{\nabla}\Phi = g\hat{\mathbf{e}}_0$, the $\pm$ components will be zero. On the other hand, the 0 component won't be zero. The gradient of $\rho\mathbf{u}$ is given by

$$\boldsymbol{\nabla} \cdot (\rho \mathbf{u}) = \sum_{lm} \left[\frac{d}{dr}(\rho u^0_{lm}) + 2r^{-1}\rho u^0_{lm} - \Omega^0_l r^{-1} \rho (u^-_{lm} + u^+_{lm}) \right] Y^0_{lm}. \tag{157}$$

The 0-component is given by

$$\int_{\mathcal{M}} (\mathbf{w}^\dagger \cdot \boldsymbol{\nabla}\Phi)[\boldsymbol{\nabla} \cdot (\rho \mathbf{u})] \, \mathrm{d}V = \int_{\mathcal{R}} \overline{w^0_{lm}} g r \left\{ r \frac{d}{dr}(\rho u^0_{lm}) + \rho \left[2u^0_{lm} - \Omega^0_l (u^-_{lm} + u^+_{lm}) \right] \right\} \mathrm{d}r \tag{158}$$

RHS

The RHS is given by

$$\int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^\dagger \, \mathrm{d}V, \tag{159}$$

where

$$\boldsymbol{\Gamma} = \mathbf{M}\delta(\mathbf{x} - \mathbf{x}_s)H(t), \tag{160}$$

with $\mathbf{M}$ as the moment tensor, $\mathbf{x}_s$ is the source location (for a point source). Now, this is identical to the weak form for the stress tensor. We could write the force tensor explicitly, but this is not the easiest way to do the calculation, due to the Dirac delta. Defining $\mathbf{G} = \boldsymbol{\nabla}\mathbf{w}$, and write out the components of $\mathbf{G}$ based on Dahlen and Tromp [4].

$\boldsymbol{\alpha} = \mathbf{0}$:

$$G^{00} = \dot{w}^0_{lm}, \tag{161}$$
$$G^{\pm 0} = r^{-1}\Omega^0_l w^0_{lm}, \tag{162}$$
$$G^{\pm\mp} = -r^{-1} w^0_{lm}. \tag{163}$$

$\boldsymbol{\alpha} = \pm$:

$$G^{0\pm} = \dot{w}^\pm_{lm}, \tag{164}$$
$$G^{\pm 0} = -r^{-1} w^\pm_{lm}, \tag{165}$$
$$G^{\pm\pm} = \Omega^2_l r^{-1} w^\pm_{lm}, \tag{166}$$
$$G^{\mp\pm} = r^{-1}\Omega^0_l w^\pm_{lm}. \tag{167}$$

The integrand is then

$$\begin{aligned} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^\dagger &= \sum_{\alpha\beta} \Gamma^{\alpha\beta} \overline{G^{\alpha\beta}}, \\ &= \sum_{\alpha\beta} M^{\alpha\beta} \overline{G^{\alpha\beta}} \delta(\mathbf{x} - \mathbf{x}_s) H(t). \end{aligned} \tag{168}$$

Consequently, the RHS is given by (ignoring the Heaviside step function):

$\boldsymbol{\alpha} = \mathbf{0}$:

$$
\begin{aligned}
&\int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^{\dagger}\,\mathrm{d}V \\
&= \sum_{\alpha\beta} \int_{\mathcal{M}} M^{\alpha\beta}\overline{G^{\alpha\beta}}\delta(\mathbf{x}-\mathbf{x}_s)\,\mathrm{d}V, \\
&= \int_{\mathcal{M}} \left\{ M^{00}\overline{\dot{w}^0_{lm}Y^0_{lm}} + M^{\pm 0}r^{-1}\Omega^0_l\overline{w^0_{lm}Y^{\pm}_{lm}} - M^{\pm\mp}r^{-1}\overline{w^0_{lm}Y^0_{lm}} \right\}\delta(\mathbf{x}-\mathbf{x}_s)\,\mathrm{d}V, \\
&= M^{00}\overline{\dot{w}^0_{lm}(r)Y^0_{lm}(\theta_s,\varphi_s)} + r_s^{-1}\overline{w^0_{lm}(r_s)}\Big\{\Omega^0_l\Big[M^{+0}\overline{Y^+_{lm}(\theta_s,\varphi_s)} + M^{-0}\overline{Y^-_{lm}(\theta_s,\varphi_s)}\Big] - \\
&\qquad\qquad - \overline{Y^0_{lm}(\theta_s,\varphi_s)}\left[M^{+-}+M^{-+}\right]\Big\}
\end{aligned}
\tag{169}
$$

$\boldsymbol{\alpha} = \pm$:

$$
\begin{aligned}
&\int_{\mathcal{M}} \boldsymbol{\Gamma} : \boldsymbol{\nabla}\mathbf{w}^{\dagger}\,\mathrm{d}V \\
&= \sum_{\alpha\beta} \int_{\mathcal{M}} M^{\alpha\beta}\overline{G^{\alpha\beta}}\delta(\mathbf{x}-\mathbf{x}_s)\,\mathrm{d}V, \\
&= \int_{\mathcal{M}} \Big\{ M^{0\pm}\overline{\dot{w}^{\pm}_{lm}Y^{\pm}_{lm}} - M^{\pm 0}r^{-1}\overline{w^{\pm}_{lm}Y^{\pm}_{lm}} + \Omega^2_l M^{\pm\pm}r^{-1}\overline{w^{\pm}_{lm}Y^{\pm\pm}_{lm}} + \\
&\qquad + M^{\mp\pm}r^{-1}\Omega^0_l\overline{w^{\pm}_{lm}Y^0_{lm}}\Big\}\delta(\mathbf{x}-\mathbf{x}_s)\,\mathrm{d}V, \\
&= M^{0\pm}\overline{\dot{w}^{\pm}_{lm}(r_s)Y^{\pm}_{lm}(\theta_s,\varphi_s)} - M^{\pm 0}r_s^{-1}\overline{w^{\pm}_{lm}(r_s)Y^{\pm}_{lm}(\theta_s,\varphi_s)} + \\
&\quad \Omega^2_l M^{\pm\pm}r_s^{-1}\overline{w^{\pm}_{lm}(r_s)Y^{\pm\pm}_{lm}(\theta_s,\varphi_s)} + M^{\mp\pm}r_s^{-1}\Omega^0_l\overline{w^{\pm}_{lm}(r_s)Y^0_{lm}(\theta_s,\varphi_s)}, \\
&= M^{0\pm}\overline{\dot{w}^{\pm}_{lm}(r_s)Y^{\pm}_{lm}(\theta_s,\varphi_s)} + \\
&\quad + r_s^{-1}\overline{w^{\pm}_{lm}(r_s)}\left[\Omega^0_l M^{\mp\pm}\overline{Y^0_{lm}(\theta_s,\varphi_s)} - M^{\pm 0}\overline{Y^{\pm}_{lm}(\theta_s,\varphi_s)} + \Omega^2_l M^{\pm\pm}\overline{Y^{\pm\pm}_{lm}(\theta_s,\varphi_s)}\right]
\end{aligned}
\tag{170}
$$

8.3.3 The weak form for the 0-component

The 0-component, ie the weak form with $\mathbf{w} = w^0_{lm}Y^0_{lm}\hat{\mathbf{e}}_0$, we can put all the terms together and find

$$
\begin{aligned}
&-\omega^2\int_{\mathcal{R}} r^2\rho\overline{w^0_{lm}}u^0_{lm}\,\mathrm{d}r + \int_{\mathcal{R}} r^2\left\{\overline{\dot{w}^0_{lm}}T^{00}_{lm} + r^{-1}w^0_{lm}\left[\Omega^0_l(T^{+0}_{lm}+T^{-0}_{lm}) - (T^{+-}_{lm}+T^{-+}_{lm})\right]\right\}\mathrm{d}r + \\
&+\int_{\mathcal{R}} r^2\left[\overline{w^0_{lm}}\cdot\frac{d}{dr}(\rho g u^0_{lm})\right]\mathrm{d}r + \int_{\mathcal{R}} r^2\rho\overline{w^0_{lm}}\dot{P}_{lm}\,\mathrm{d}r - \\
&-\int_{\mathcal{R}}\overline{w^0_{lm}}gr\left\{r\frac{d}{dr}(\rho u^0_{lm}) + \rho\left[2u^0_{lm} - \Omega^0_l(u^-_{lm}+u^+_{lm})\right]\right\}\mathrm{d}r = \\
&= H(\omega)M^{00}\overline{\dot{w}^0_{lm}(r)Y^0_{lm}(\theta_s,\varphi_s)} + \\
&+ H(\omega)r_s^{-1}\overline{w^0_{lm}(r_s)}\Big\{\Omega^0_l\Big[M^{+0}\overline{Y^+_{lm}(\theta_s,\varphi_s)} + M^{-0}\overline{Y^-_{lm}(\theta_s,\varphi_s)}\Big] - \overline{Y^0_{lm}(\theta_s,\varphi_s)}\left[M^{+-}+M^{-+}\right]\Big\}.
\end{aligned}
\tag{171}
$$

To aid with the comprehensibility of this equation we will drop (l, m) subscript, but not the α superscript. Employing (105) gives

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w^0} u^0 \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \left\{ \overline{\dot{w}^0} T^{00} + r^{-1} w^0 \left[\Omega_l^0 (T^{+0} + T^{-0}) - 2T^{-+} \right] \right\} \mathrm{d}r + \\
&+ \int_{\mathcal{R}} r^2 \left[\overline{w^0} \cdot \frac{d}{dr}(\rho g u^0) \right] \mathrm{d}r + \int_{\mathcal{R}} r^2 \rho \overline{w^0} \dot{P} \,\mathrm{d}r - \\
&- \int_{\mathcal{R}} \overline{w^0} g r \left\{ r \frac{d}{dr}(\rho u^0) + \rho \left[2u^0 - \Omega_l^0 (u^- + u^+) \right] \right\} \mathrm{d}r = \\
&= H(\omega) M^{00} \overline{\dot{w}^0(r) Y^0(\theta_s, \varphi_s)} + \\
&+ H(\omega) r_s^{-1} \overline{w^0(r_s)} \left\{ \Omega_l^0 \left[M^{+0} \overline{Y^+(\theta_s, \varphi_s)} + M^{-0} \overline{Y^-(\theta_s, \varphi_s)} \right] - \overline{Y^0(\theta_s, \varphi_s)} \left[M^{+-} + M^{-+} \right] \right\}. \qquad (172)
\end{aligned}
$$

It is possible to simplify this expression further. In particular we use

$$
\begin{aligned}
\frac{d}{dr}(\rho g u^0) &= g \partial_r (\rho u^0) + \rho \dot{g} u^0, \\
&= g \partial_r (\rho u^0) + 4\pi G \rho^2 u^0 - 2r^{-1} g \rho u^0. \qquad (173)
\end{aligned}
$$

The terms in (172) involving these derivatives are

$$
\begin{aligned}
&\int_{\mathcal{R}} r^2 \left[\overline{w^0} \cdot \frac{d}{dr}(\rho g u^0) \right] \mathrm{d}r - \int_{\mathcal{R}} \overline{w^0} g r \cdot r \frac{d}{dr}(\rho u^0) \,\mathrm{d}r, \\
&= \int_{\mathcal{R}} r^2 \overline{w^0} [g \partial_r (\rho u^0) + 4\pi G \rho^2 u^0 - 2r^{-1} g \rho u^0] \,\mathrm{d}r - \int_{\mathcal{R}} \overline{w^0} g r^2 \partial_r (\rho u^0) \,\mathrm{d}r, \\
&= 4\pi G \int_{\mathcal{R}} r^2 \rho^2 \overline{w^0} u^0 \,\mathrm{d}r - 2 \int_{\mathcal{R}} r g \rho \overline{w^0} u^0 \,\mathrm{d}r. \qquad (174)
\end{aligned}
$$

Thus we have

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} \rho r^2 \overline{w^0} u^0 \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{\dot{w}^0} T^{00} \,\mathrm{d}r + \int_{\mathcal{R}} r w^0 \left[\Omega_l^0 (T^{+0} + T^{-0}) - 2T^{-+} \right] \mathrm{d}r \\
&+ \int_{\mathcal{R}} r \rho \overline{w^0} \left\{ r \dot{P} + (4\pi G \rho r - 4g) u^0 + \Omega_l^0 g (u^- + u^+) \right\} \mathrm{d}r \\
&= H(\omega) M^{00} \overline{\dot{w}^0(r) Y^0(\theta_s, \varphi_s)} + \\
&+ H(\omega) r_s^{-1} \overline{w^0(r_s)} \left\{ \Omega_l^0 \left[M^{+0} \overline{Y^+(\theta_s, \varphi_s)} + M^{-0} \overline{Y^-(\theta_s, \varphi_s)} \right] - \overline{Y^0(\theta_s, \varphi_s)} \left[M^{+-} + M^{-+} \right] \right\}. \qquad (175)
\end{aligned}
$$

It is easy to see that the left hand side is the equivalent weak form to equation (8.74) of Dahlen and Tromp [4].

8.3.4 The weak form for the other components

The $\pm$-components, ie with $\mathbf{w} = w^{\pm}_{lm} Y^{\pm}_{lm} \hat{\mathbf{e}}_{\pm}$, are given by

$$
\begin{aligned}
&- \omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w^{\pm}_{lm}} u^{\pm}_{lm} \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \left\{ \overline{\dot{w}^{\pm}_{lm}} T^{0\pm}_{lm} + r^{-1} \overline{w^{\pm}_{lm}} \left[\Omega^2_l T^{\pm\pm} + \Omega^0_l T^{\mp\pm} - T^{\pm 0}_{lm} \right] \right\} \mathrm{d}r \\
&+ \Omega^0_l \int_{\mathcal{R}} \rho g r \overline{w^{\pm}_{lm}} u^0_{lm} \,\mathrm{d}r + \Omega^0_l \int_{\mathcal{R}} r \rho \overline{w^{\pm}_{lm}} P_{lm} \,\mathrm{d}r \\
&= H(\omega) M^{0\pm} \overline{\dot{w}^{\pm}_{lm}(r_s) Y^{\pm}_{lm}(\theta_s, \varphi_s)} + \\
&\quad + H(\omega) r_s^{-1} \overline{w^{\pm}_{lm}(r_s)} \left[\Omega^0_l M^{\mp\pm} \overline{Y^0_{lm}(\theta_s, \varphi_s)} - M^{\pm 0} \overline{Y^{\pm}_{lm}(\theta_s, \varphi_s)} + \Omega^2_l M^{\pm\pm} \overline{Y^{\pm\pm}_{lm}(\theta_s, \varphi_s)} \right].
\end{aligned} \tag{176}
$$

Using (105), (104), and dropping subscript lm

$$
\begin{aligned}
&- \omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w^{\pm}} u^{\pm} \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{\dot{w}^{\pm}} T^{0\pm} \,\mathrm{d}r + \int_{\mathcal{R}} r \overline{w^{\pm}} \left\{ \Omega^0_l \left[T^{-+} + \rho(P + g u^0) \right] + \Omega^2_l T^{\pm\pm} - T^{0\pm} \right\} \mathrm{d}r \\
&= H(\omega) M^{0\pm} \overline{\dot{w}^{\pm}(r_s) Y^{\pm}(\theta_s, \varphi_s)} + \\
&\quad + H(\omega) r_s^{-1} \overline{w^{\pm}(r_s)} \left[\Omega^0_l M^{\mp\pm} \overline{Y^0(\theta_s, \varphi_s)} - M^{\pm 0} \overline{Y^{\pm}(\theta_s, \varphi_s)} + \Omega^2_l M^{\pm\pm} \overline{Y^{\pm\pm}(\theta_s, \varphi_s)} \right].
\end{aligned} \tag{177}
$$

8.3.5 Weak form for gravity problem

In addition to the three components of the displacement we also need a weak form for the gravitational perturbation ϕ. The strong form for ϕ is given by

$$
\nabla^2 \phi = -4\pi G \boldsymbol{\nabla} \cdot (\rho \mathbf{u}), \tag{178}
$$

The boundary conditions are

$$
[\phi]^+_- = 0, \quad x \in \Sigma, \tag{179}
$$

$$
\left[(4\pi G)^{-1} \hat{\mathbf{n}} \cdot \nabla \phi \right]^+_- - \rho^- \hat{\mathbf{n}} \cdot \mathbf{u}^- = 0, \quad x \in \partial M, \tag{180}
$$

$$
\left[(4\pi G)^{-1} \hat{\mathbf{n}} \cdot \nabla \phi + \rho \hat{\mathbf{n}} \cdot \mathbf{u} \right]^+_- = 0, \quad x \in \Sigma_{SS}, \tag{181}
$$

$$
\left[(4\pi G)^{-1} \hat{\mathbf{n}} \cdot \nabla \phi \right]^+_- + [\rho]^+_- \, \hat{\mathbf{n}} \cdot \mathbf{u}^+ = 0, \quad x \in \Sigma_{FS}, \tag{182}
$$

$$
\left[(4\pi G)^{-1} \hat{\mathbf{n}} \cdot \nabla \phi \right]^+_- + [\rho]^+_- \, \hat{\mathbf{n}} \cdot \mathbf{u}^- = 0, \quad x \in \Sigma_{SF}. \tag{183}
$$

Now, it is useful to find the divergence term before getting the weak form:

$$
\begin{aligned}
\boldsymbol{\nabla} \cdot (\rho \mathbf{u}) &= \boldsymbol{\nabla} \cdot \left\{ \sum_{lm} \left[\rho(r) u^-_{lm} Y^-_{lm} \hat{\mathbf{e}}_- + \rho(r) u^0_{lm} Y^0_{lm} \hat{\mathbf{e}}_0 + \rho(r) u^+_{lm} Y^+_{lm} \hat{\mathbf{e}}_+ \right] \right\}, \\
&= \sum_{lm} \left\{ \partial_r [\rho(r) u^0_{lm}(r)] + 2 r^{-1} \rho(r) u^0_{lm}(r) - \Omega^0_l r^{-1} \rho(r) (u^-_{lm}(r) + u^+_{lm}(r)) \right\} Y^0_{lm}.
\end{aligned} \tag{184}
$$

To get the weak form of (178) we multiply by a test function $\overline{\psi}$ and integrate over the body

$$
\begin{aligned}
\int_{\mathcal{M}} \overline{\psi} \nabla^2 \phi \,\mathrm{d}V &= -4\pi G \int_{\mathcal{M}} \overline{\psi} \boldsymbol{\nabla} \cdot (\rho \mathbf{u}) \,\mathrm{d}V, \\
\int_{\Sigma} \overline{\psi} \hat{\mathbf{n}} \cdot \boldsymbol{\nabla} \phi \,\mathrm{d}S - \int_{\mathcal{M}} \boldsymbol{\nabla} \overline{\psi} \cdot \boldsymbol{\nabla} \phi \,\mathrm{d}V &= -4\pi G \int_{\Sigma} \overline{\psi} \rho \hat{\mathbf{n}} \cdot \mathbf{u} \,\mathrm{d}S + 4\pi G \int_{\mathcal{M}} \overline{\boldsymbol{\nabla} \psi} \cdot \rho \mathbf{u} \,\mathrm{d}V, \\
- \int_{\mathcal{M}} \boldsymbol{\nabla} \overline{\psi} \cdot \boldsymbol{\nabla} \phi \,\mathrm{d}V + \int_{\Sigma} \overline{\psi} [\boldsymbol{\nabla} \phi + 4\pi G \rho \mathbf{u}] \cdot \hat{\mathbf{n}} \,\mathrm{d}S &= 4\pi G \int_{\mathcal{M}} \overline{\boldsymbol{\nabla} \psi} \cdot \rho \mathbf{u} \,\mathrm{d}V.
\end{aligned} \tag{185}
$$

Now, the boundary condition for the gravitational perturbation is $\hat{\mathbf{n}} \cdot [\nabla\phi + 4\pi G\rho\mathbf{u}]_-^+ = 0$ (this is the general case, the BCs from Crawford et al. [3] arise from considering the potential in the fluid regions). The surface integral will cancel on all internal boundaries, and on the external boundary we are left with the derivative outside the planet, ie

$$-\int_{\mathcal{M}} \boldsymbol{\nabla}\overline{\psi} \cdot \boldsymbol{\nabla}\phi \,\mathrm{d}V + \int_{\partial\mathcal{M}} \overline{\psi}\partial_r\phi|_+ \,\mathrm{d}S = 4\pi G \int_{\mathcal{M}} \overline{\boldsymbol{\nabla}\psi} \cdot \rho\mathbf{u} \,\mathrm{d}V, \tag{186}$$

where $\partial_r\phi|_+$ indicates the radial derivative of the exterior potential on the surface of the planet. Using the exterior representation

$$\phi(r > b) = \sum_{lm} \phi_{lm} \left(\frac{b}{r}\right)^{l+1} Y^0_{lm}(\theta, \varphi). \tag{187}$$

This implies that the partial derivative on the surface is given by

$$\partial_r\phi = -b^{-1} \sum_{lm} (l+1)\phi_{lm} Y^0_{lm}. \tag{188}$$

Using this result,

$$\boldsymbol{\nabla}\phi = \sum_{lm} [\partial_r\phi_{lm} Y^0_{lm}\hat{\mathbf{e}}_0 + \Omega^0_l r^{-1}\phi_{lm}(Y^+_{lm}\hat{\mathbf{e}}_+ + Y^-_{lm}\hat{\mathbf{e}}_-)], \tag{189}$$

and taking the test function $\psi = \psi_{lm} Y^0_{lm}$ we arrive at the weak form

$$\int_{\mathcal{R}} [r^2(\partial_r\overline{\psi})(\partial_r\phi) + 2(\Omega^0_l)^2\overline{\psi}\phi] \,\mathrm{d}r + b(l+1)\overline{\psi}\phi = -4\pi G \int_{\mathcal{R}} \rho(r)[r^2(\partial_r\overline{\psi})u^0 + \Omega^0_l r\overline{\psi}(u^+ + u^-)] \,\mathrm{d}r, \tag{190}$$

where for ease we have dropped the (lm) subscript.

Strong form equivalency We show equivalency with the strong form in Dahlen and Tromp [4]. In particular, equation (8.76) is

$$\ddot{P} + 2r^{-1}\dot{P} - l(l+1)r^{-2}P = -4\pi G\dot{\rho}u^0 - 4\pi G\rho[\dot{u}^0 + 2r^{-1}u^0 - \Omega^0_l r^{-1}(u^- + u^+)], \tag{191}$$

subject to $[P]_-^+ = 0$ and $[\dot{P} + 4\pi G\rho u^0]_-^+ = 0$ on discontinuities. Multiplying by $r^2\tilde{P}$, with $\tilde{P}$ as our test function, we obtain

$$\frac{d}{dr}(r^2\tilde{P}\dot{P}) - r^2\dot{\tilde{P}}\dot{P} - k^2\tilde{P}P = -4\pi G\frac{d}{dr}(\rho r^2\tilde{P}u^0) + 4\pi G\rho\left[u^0\frac{d}{dr}(r^2\tilde{P}) - r\tilde{P}(2u^0 - \Omega^0_l(u^- + u^+))\right], \tag{192}$$

which can be rewritten as

$$\frac{d}{dr}\left[r^2\tilde{P}(\dot{P} + 4\pi G\rho u^0)\right] - r^2\dot{\tilde{P}}\dot{P} - k^2\tilde{P}P = 4\pi G\rho\left[r^2u^0\tilde{P} + r\Omega^0_l\tilde{P}(u^- + u^+)\right]. \tag{193}$$

Integrating over $\mathcal{R}$, we can apply the boundary conditions for P on internal boundaries, eliminating these terms. Again, on the outer boundary we have $\dot{P}_- + 4\pi G\rho_- u^0(b_-) = \dot{P}_+$. Using the external representation of the potential we once again obtain

$$\int_{\mathcal{R}} \left[r^2(\partial_r\tilde{P})(\partial_r P) + k^2\tilde{P}P\right] \mathrm{d}r + b(l+1)\tilde{P}P = -4\pi G \int_{\mathcal{R}} \rho(r)\left[r^2(\partial_r\tilde{P})u^0 + \Omega^0_l r\tilde{P}(u^- + u^+)\right] \mathrm{d}r. \tag{194}$$

Finally, by their definitions $k^2 = 2(\Omega^0_l)^2$. Consequently the two equations are the same.

8.4 Toroidal free oscillations

8.4.1 The weak form

If we take (177) with the positive component and negative component and subtract one from the other, set $w^- = w^+$, and assume that is a symmetric moment tensor then we find that $u^- - u^+$ must satisfy:

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w}(u^- - u^+)\,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{w}(T^{0-} - T^{0+})\,\mathrm{d}r + \int_{\mathcal{R}} r\overline{w}\left\{\Omega_l^2(T^{--} - T^{++}) - (T^{0-} - T^{0+})\right\}\mathrm{d}r \\
&= H(\omega)\left[\overline{\dot{w}(r_s)} - r_s^{-1}\overline{w(r_s)}\right]\left[M^{0-}\overline{Y^-(\theta_s,\varphi_s)} - M^{0+}\overline{Y^+(\theta_s,\varphi_s)}\right] + \\
&\quad + H(\omega)\Omega_l^2 r_s^{-1}\overline{w(r_s)}\left[M^{--}\overline{Y^{--}(\theta_s,\varphi_s)} - M^{++}\overline{Y^{++}(\theta_s,\varphi_s)}\right].
\end{aligned} \tag{195}
$$

Using $W = \frac{i}{\sqrt{2}}(u^- - u^+)$, we have

$$
T^{0-} - T^{0+} = \mu[(\dot{u}^- - \dot{u}^+) - r^{-1}(u^- - u^+)], \tag{196}
$$

$$
T^{--} - T^{++} = 2\Omega_l^2 \mu r^{-1}(u^- - u^+). \tag{197}
$$

The traction is given by $T = \frac{i}{\sqrt{2}}(T^{0-} - T^{0+}) = \mu[\dot{W} - r^{-1}W]$. Rewriting $w \to \tilde{W}$, dropping overbars and using $\Omega_l^{\pm N} = \sqrt{\frac{1}{2}(l \pm N)(l \mp N + 1)}$:

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \tilde{W} W\,\mathrm{d}r + \int_{\mathcal{R}} r^2 \partial_r \tilde{W} T\,\mathrm{d}r + \int_{\mathcal{R}} r\tilde{W}\left\{2(\Omega_l^2)^2 \mu r^{-1} W - T\right\}\mathrm{d}r \\
&= -\omega^2 \int_{\mathcal{R}} r^2 \rho \tilde{W} W\,\mathrm{d}r + \int_{\mathcal{R}} r[r(\partial_r \tilde{W}) - \tilde{W}]T\,\mathrm{d}r + (l+2)(l-1)\int_{\mathcal{R}} \mu \tilde{W} W\,\mathrm{d}r, \\
&= -\omega^2 \int_{\mathcal{R}} \rho \tilde{W} W r^2\,\mathrm{d}r + \int_{\mathcal{R}} \mu[r\partial_r \tilde{W} - \tilde{W}][r\partial_r W - W]\,\mathrm{d}r + (k^2 - 2)\int_{\mathcal{R}} \mu \tilde{W} W\,\mathrm{d}r.
\end{aligned} \tag{198}
$$

The forced problem is given by

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} \rho \tilde{W} W r^2\,\mathrm{d}r + \int_{\mathcal{R}} \mu[r\partial_r \tilde{W} - \tilde{W}][r\partial_r W - W]\,\mathrm{d}r + (k^2 - 2)\int_{\mathcal{R}} \mu \tilde{W} W\,\mathrm{d}r \\
&= H(\omega)\frac{i}{\sqrt{2}}\left[(\partial_r \tilde{W})(r_s) - r_s^{-1}\tilde{W}\right]\left[M^{0-}\overline{Y^-(\theta_s,\varphi_s)} - M^{0+}\overline{Y^+(\theta_s,\varphi_s)}\right] \\
&\quad + H(\omega)\frac{i}{\sqrt{2}}\Omega_l^2 r_s^{-1}\tilde{W}(r_s)\left[M^{--}\overline{Y^{--}(\theta_s,\varphi_s)} - M^{++}\overline{Y^{++}(\theta_s,\varphi_s)}\right].
\end{aligned} \tag{199}
$$

The relevant moment tensor components are

$$
M^{0\pm} = \frac{1}{\sqrt{2}}(\mp M_{r\theta} + iM_{r\varphi}), \tag{200}
$$

$$
M^{\pm\pm} = \frac{1}{2}(M_{\theta\theta} - M_{\varphi\varphi} \mp 2iM_{\theta\varphi}). \tag{201}
$$

Finally, to conclude this short theory section we show that the forced problem satisfies the criteria on W_{lm} (which is more a check for myself). It should be noted that we do not include $H(\omega)$ in this proof, because the frequency domain relations are slightly different. The test functions $\tilde{W}$ are taken

to be real. We use $\overline{M^{0-}} = -M^{0+}$ and $\overline{M^{--}} = M^{++}$ and take the conjugate of the relevant parts of the right hand side, ie

$$\begin{aligned}
\mathrm{RRHS}_{lm} &= i\left[M^{0-}\overline{Y^{-}_{lm}} - M^{0+}\overline{Y^{+}_{lm}}\right] + i\left[M^{--}\overline{Y^{--}_{lm}} - M^{++}\overline{Y^{++}_{lm}}\right], \\
\overline{\mathrm{RRHS}_{lm}} &= -i\left[-M^{0+}Y^{-}_{lm} + M^{0-}Y^{+}_{lm}\right] - i\left[M^{++}Y^{--}_{lm} - M^{--}Y^{++}_{lm}\right], \\
&= i\left[M^{0+}(-1)^{m+1}\overline{Y^{+}_{l,-m}} - M^{0-}(-1)^{m+1}\overline{Y^{-}_{l,-m}}\right] \\
&\qquad + i\left[M^{--}(-1)^{m+2}\overline{Y^{--}_{l,-m}} - M^{++}(-1)^{m+2}\overline{Y^{++}_{l,-m}}\right], \\
&= (-1)^{m}\mathrm{RRHS}_{l,-m}.
\end{aligned} \tag{202}$$

The forcing terms satisfy the relation for W_{lm}.

Thus we need to solve

$$\begin{aligned}
&-\omega^2\int_{\mathcal{R}}\rho\tilde{W}Wr^2\,\mathrm{d}r + \int_{\mathcal{R}}\mu[r\partial_r\tilde{W} - \tilde{W}][r\partial_r W - W]\,\mathrm{d}r + (k^2-2)\int_{\mathcal{R}}\mu\tilde{W}W\,\mathrm{d}r \\
&= \frac{1}{\omega\sqrt{2}}\Bigg\{\left[(\partial_r\tilde{W})(r_s) - r_s^{-1}\tilde{W}\right]\left[M^{0-}\overline{Y^{-}(\theta_s,\varphi_s)} - M^{0+}\overline{Y^{+}(\theta_s,\varphi_s)}\right] \\
&\quad + \Omega_l^2 r_s^{-1}\tilde{W}(r_s)\left[M^{--}\overline{Y^{--}(\theta_s,\varphi_s)} - M^{++}\overline{Y^{++}(\theta_s,\varphi_s)}\right]\Bigg\}.
\end{aligned} \tag{203}$$

8.4.2 Ground motion

Ultimately we want the ground motion at a particular location. The toroidal free oscillation is given by

$$\mathbf{w} = \frac{i}{\sqrt{2}}\sum_{lm} W_{lm}\left[Y^{+}_{lm}\hat{\mathbf{e}}_{+} - Y^{-}_{lm}\hat{\mathbf{e}}_{-}\right]. \tag{204}$$

Thus we have

$$\mathbf{w}(\theta_r,\varphi_r) = \frac{i}{\sqrt{2}}\sum_{lm} W_{lm}(r_r)\left[Y^{+}_{lm}(\theta_r,\varphi_r)\hat{\mathbf{e}}_{+}(\theta_r,\varphi_r) - Y^{-}_{lm}(\theta_r,\varphi_r)\hat{\mathbf{e}}_{-}(\theta_r,\varphi_r)\right]. \tag{205}$$

The receiver vector is a set of coefficients or functions that, when applied to the modal expansion of the displacement field, yield the ground motion at a specific receiver location (e.g., at the Earth's surface at coordinates (r_r,θ_r,φ_r)). For the toroidal component, the receiver vector $\mathbf{R}$ at (r_r,θ_r,φ_r) is defined by evaluating the basis functions at the receiver location:

$$\mathbf{R}_{lm}(r_r,\theta_r,\varphi_r) = \frac{i}{\sqrt{2}}\left[Y^{+}_{lm}(\theta_r,\varphi_r)\,\hat{\mathbf{e}}_{+}(\theta_r,\varphi_r) - Y^{-}_{lm}(\theta_r,\varphi_r)\,\hat{\mathbf{e}}_{-}(\theta_r,\varphi_r)\right] \tag{206}$$

The ground motion at the receiver is then given by

$$\mathbf{w}(r_r,\theta_r,\varphi_r) = \sum_{lm} W_{lm}(r_r)\,\mathbf{R}_{lm}(r_r,\theta_r,\varphi_r), \tag{207}$$

where $W_{lm}(r_r)$ are the modal amplitudes evaluated at the receiver radius.

The θ and φ components of the ground motion at the receiver are obtained by projecting $\mathbf{w}(r_r, \theta_r, \varphi_r)$ onto the local unit vectors $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\varphi}}$. Using the definitions of the canonical basis vectors:

$$\hat{\mathbf{e}}_+ = -\frac{1}{\sqrt{2}}(\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\varphi}}), \tag{208}$$

$$\hat{\mathbf{e}}_- = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{\theta}} - i\hat{\boldsymbol{\varphi}}), \tag{209}$$

the projections are:

$$\begin{aligned} w_\theta(r_r, \theta_r, \varphi_r) &= \mathbf{w}(r_r, \theta_r, \varphi_r) \cdot \hat{\boldsymbol{\theta}} \\ &= \frac{i}{\sqrt{2}} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) \left(-\frac{1}{\sqrt{2}} \right) - Y^-_{lm}(\theta_r, \varphi_r) \left(\frac{1}{\sqrt{2}} \right) \right] \\ &= -\frac{i}{2} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) + Y^-_{lm}(\theta_r, \varphi_r) \right], \end{aligned} \tag{210}$$

$$\begin{aligned} w_\varphi(r_r, \theta_r, \varphi_r) &= \mathbf{w}(r_r, \theta_r, \varphi_r) \cdot \hat{\boldsymbol{\varphi}} \\ &= \frac{i}{\sqrt{2}} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) \left(-\frac{i}{\sqrt{2}} \right) - Y^-_{lm}(\theta_r, \varphi_r) \left(-\frac{i}{\sqrt{2}} \right) \right] \\ &= \frac{1}{2} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) - Y^-_{lm}(\theta_r, \varphi_r) \right]. \end{aligned} \tag{211}$$

Thus, the θ and φ components of the toroidal ground motion at the receiver are:

$$w_\theta(r_r, \theta_r, \varphi_r) = -\frac{i}{2} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) + Y^-_{lm}(\theta_r, \varphi_r) \right], \tag{212}$$

$$w_\varphi(r_r, \theta_r, \varphi_r) = \frac{1}{2} \sum_{lm} W_{lm}(r_r) \left[Y^+_{lm}(\theta_r, \varphi_r) - Y^-_{lm}(\theta_r, \varphi_r) \right]. \tag{213}$$

8.5 The toroidal problem numerically

At this stage we do not worry excessively about computational cost. We simply assemble the full matrix and solve.

I have implemented the following:

- Full matrix calculated and then find modes
- Different matrix for each l, calculate modes for these matrices. This has the advantage of being significantly cheaper and also as the radial functions are independent of m it means that one doesn't need to worry about degeneracy of eigenfunctions of different m
- Implemented normal mode coupling of said eigenfunctions in the case of a spherically symmetric model. Currently, I
  - Calculate the modes
  - Find all the modes up to a particular frequency

  - Get them in ascending order by frequency
  - Calculate the kinetic energy and inertia matrices

- Changing the IDSM method to a new library. At this stage I have the following operability:
  - Take in complex Eigen matrices a_0, a_1, a_2 as well as complex matrix of receivers V_R and complex source vector V_S
  - Calculate the raw spectrum using either a low memory method, ie not forming $A = -\omega^2 a_2 + \omega a_1 + a_0$, or calculate with forming A. Pre-conditioning is using the block method. However, at this stage the pre-conditioner probably doesn't work except for diagonal due to changed problem

- Calculate the augmentation. As for different l the matrices are different this does it separately for the different l.

To do:

- **Augmentation:**
  - Include the augmentation functions in the coupling matrices
  - I need to figure out the order.
  - This will probably add a fair bit of cost as you need to include all augmentation matrices, so for each l, that is one per layer (plus one)

- **IDSM library:**
  - Figure out block pre-conditioning in the new format
  - Add in templating for low memory option whereby you can input operators that are matrix replacements as well as a preconditioner of another type

8.6 The problem

8.6.1 Outline

We have the following comparison between the spectral element solution and the solution using "normal mode coupling", see Fig. 15. The issue is that in the naive implementation all overtones contribute, as there is no damping of higher order modes. To do the benchmarking between the two methods we need to project the force vector in the spectral element method onto the normal modes that have been used.

In the spectral element basis we have

$$[-\omega^2 M + K]\mathbf{v}(\omega) = \mathbf{f}(\omega). \tag{214}$$

The normal mode basis is given by the solution $\omega_i^2 M\mathbf{w}_i = K\mathbf{w}_i$. The inertia and kinetic energy matrices are symmetric. Consequently we have the orthogonality condition $\mathbf{w}_i^T M\mathbf{w}_j = \delta_{ij}$. If we express the vector $\mathbf{v}$ in terms of the normal modes we have

$$\sum_j \mathbf{w}_i^T[-\omega^2 M + K]v_j(\omega)\mathbf{w}_j = \mathbf{w}_i^T\mathbf{f}(\omega). \tag{215}$$

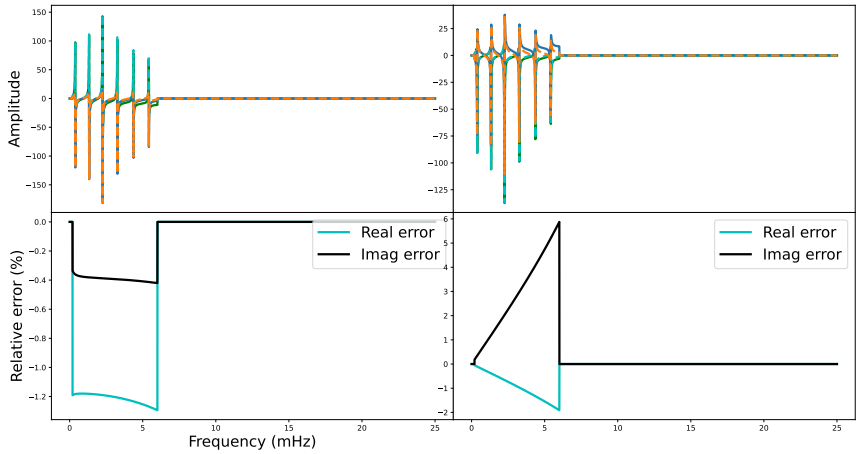

Figure 15: Comparison of spectral element solution and "normal mode coupling" solution using naive forces.

If we write $\tilde{M}_{ij} = \mathbf{w}_i^T M \mathbf{w}_j$, similarly for K, v and f we have

$$[-\omega^2 \tilde{M} + \tilde{K}]\tilde{\mathbf{v}}(\omega) = \tilde{\mathbf{f}}. \tag{216}$$

The key thing to benchmark is that in going from the normal mode to the spectral element solution we need to only include the contributions from the normal modes that we have included. It is trivial to show that this is given by

$$\mathbf{f} = \sum_i f_i M \mathbf{w}_i, \tag{217}$$

where

$$f_i = \frac{\mathbf{w}_i^T \mathbf{f}}{\mathbf{w}_i^T M \mathbf{w}_i}. \tag{218}$$

The inclusion of M ensures that the force vector is a projection onto those modes we include in the coupling. The results of performing a benchmark using this technique are shown in Fig. 16

Now, the above analysis may seem obvious. It is however a good check to do nonetheless.

8.6.2 Analysis

We have the equation

$$[-\omega^2 \tilde{M} + \tilde{K}]\tilde{\mathbf{v}}(\omega) = \tilde{\mathbf{f}}. \tag{219}$$

Now, the matrices $\tilde{M}$ and $\tilde{K}$ are diagonal. Thus, the solution is given by

$$\tilde{v}_i(\omega) = \tilde{f}_i/(-\omega^2 \tilde{M}_{ii} + \tilde{K}_{ii}). \tag{220}$$

Ultimately we therefore have

$$a(\omega) = \sum_i \frac{a_i f_i}{-\omega^2 M_{ii} + K_{ii}}. \tag{221}$$

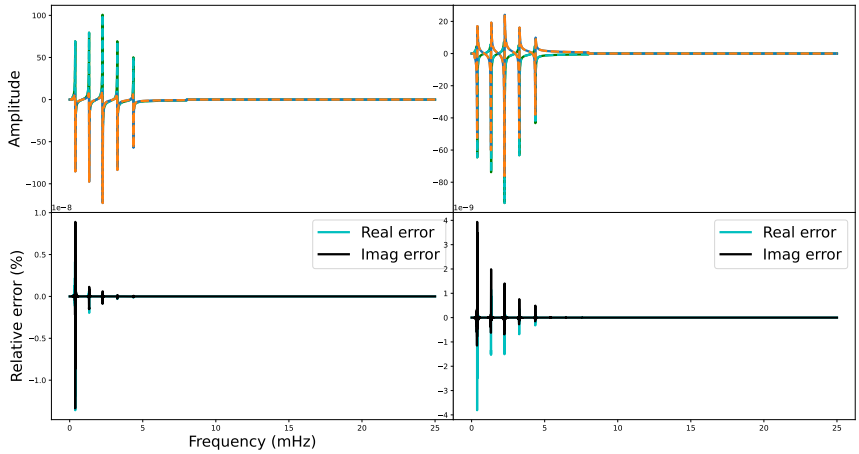

Figure 16: Comparison of spectral element solution and "normal mode coupling" solution using projection onto modes.

8.6.3 Resolution

If we process the raw output of the simulation as we would process seismic data, it turns out a lot of these issues go away. For this, we:

- Take a filter of the raw spectra in the frequency window that we want
- Transform into the time domain
- Take a filter in the time domain
- Transform back to the frequency domain

This process was originally intended to provide the most accurate method for estimating normal mode centre frequencies, by Dahlen in 1981. When we perform the filtering we find that the differences between the spectral element and normal mode solutions nearly disappear.

It should also be noted that for a deeper Earthquake we would expect less excitation of the overtones. For instance the Bolivia Earthquake of 1994. In this example we see a very clear correlation, even in the case of the raw spectra, whereby increasing the number of modes significantly improves the fit of the two methods, converging rapidly beyond a certain point.

8.7 Spheroidal free oscillations

8.7.1 The weak form

The toroidal oscillations were found by taking $W = u^- - u^+$. This was found to be independent of other combinations of the canonical components. We now need to consider the other components,

in other words $U = u^0$, $V = (u^- + u^+)/\sqrt{2}$ and P. We have the equation for the 0 component

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} \rho r^2 \overline{w^0} u^0 \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{\dot{w}^0} T^{00} \,\mathrm{d}r + \int_{\mathcal{R}} r w^0 \left[\Omega_l^0 (T^{+0} + T^{-0}) - 2T^{-+}\right] \mathrm{d}r \\
&+ \int_{\mathcal{R}} r\rho \overline{w^0} \left\{ r\dot{P} + (4\pi G\rho r - 4g) u^0 + \Omega_l^0 g (u^- + u^+) \right\} \mathrm{d}r \\
&= H(\omega) M^{00} \overline{\dot{w}^0(r) Y^0(\theta_s, \varphi_s)} + \\
&+ H(\omega) r_s^{-1} \overline{w^0(r_s)} \left\{ \Omega_l^0 \left[M^{+0} \overline{Y^+(\theta_s, \varphi_s)} + M^{-0} \overline{Y^-(\theta_s, \varphi_s)} \right] - \overline{Y^0(\theta_s, \varphi_s)} \left[M^{+-} + M^{-+} \right] \right\},
\end{aligned} \tag{222}
$$

for the $\pm$ compnents

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \overline{w^\pm} u^\pm \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{\dot{w}^\pm} T^{0\pm} \,\mathrm{d}r + \int_{\mathcal{R}} r \overline{w^\pm} \left\{ \Omega_l^0 \left[T^{-+} + \rho(P + g u^0) \right] + \Omega_l^2 T^{\pm\pm} - T^{0\pm} \right\} \mathrm{d}r \\
&= M^{0\pm} \overline{\dot{w}^\pm(r_s) Y^\pm(\theta_s, \varphi_s)} + \\
&\quad + r_s^{-1} \overline{w^\pm(r_s)} \left[\Omega_l^0 M^{\mp\pm} \overline{Y^0(\theta_s, \varphi_s)} - M^{\pm 0} \overline{Y^\pm(\theta_s, \varphi_s)} + \Omega_l^2 M^{\pm\pm} \overline{Y^{\pm\pm}(\theta_s, \varphi_s)} \right],
\end{aligned} \tag{223}
$$

and for gravity

$$
\int_{\mathcal{R}} \left[r^2 (\partial_r \tilde{P})(\partial_r P) + k^2 \tilde{P} P \right] \mathrm{d}r + b(l+1)\tilde{P}P + 4\pi G \int_{\mathcal{R}} \rho(r) \left[r^2 (\partial_r \tilde{P}) u^0 + \Omega_l^0 r \tilde{P} (u^- + u^+) \right] \mathrm{d}r = 0. \tag{224}
$$

The definitions of the traction components are

$$
\begin{aligned}
T^{00} &= (\kappa + \tfrac{4}{3}\mu) \dot{u}_{lm}^0 + (\kappa - \tfrac{2}{3}\mu) r^{-1} [2u_{lm}^0 - \Omega_l^0 (u_{lm}^- + u_{lm}^+)], && (225) \\
T^{\pm\pm} &= 2\mu \Omega_l^2 r^{-1} u_{lm}^\pm, && (226) \\
T^{\pm 0} &= T^{0\pm} = \mu \left[\dot{u}_{lm}^\pm - r^{-1} (u_{lm}^\pm - \Omega_l^0 u_{lm}^0) \right], && (227) \\
T^{\pm\mp} &= -(\kappa - \tfrac{2}{3}\mu) \dot{u}_{lm}^0 - (\kappa + \tfrac{1}{3}\mu) r^{-1} \left[2u_{lm}^0 - \Omega_l^0 (u_{lm}^- + u_{lm}^+) \right]. && (228)
\end{aligned}
$$

Writing $S = (T^{0-} + T^{0+})/\sqrt{2}$, we have

$$
\begin{aligned}
S &= \frac{\mu}{\sqrt{2}} [\dot{u}^- + \dot{u}^+ - r^{-1}(u^+ + u^- - 2\Omega_l^0 u^0)], \\
&= \mu [\dot{V} - r^{-1}(V - \sqrt{2}\Omega_l^0 U)], \\
&= \mu [\dot{V} + k r^{-1} U - r^{-1} V].
\end{aligned} \tag{229}
$$

Similarly, we have

$$
(T^{--} + T^{++})/\sqrt{2} = 2\mu \Omega_l^2 r^{-1} V, \tag{230}
$$

$$
T^{-+} = -(\kappa - \tfrac{2}{3}\mu) \dot{U} - (\kappa + \tfrac{1}{3}\mu) r^{-1} \left[2U - kV \right] \tag{231}
$$

Left hand side

Substituting into (222) we have for the left hand side (writing test function with a prime)

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} \rho r^2 \overline{w^0} u^0 \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \overline{\dot{w}^0} T^{00} \,\mathrm{d}r + \int_{\mathcal{R}} r w^0 \left[\Omega_l^0 (T^{+0} + T^{-0}) - 2T^{-+}\right] \mathrm{d}r \\
&+ \int_{\mathcal{R}} r\rho \overline{w^0} \left\{ r\dot{P} + (4\pi G\rho r - 4g) u^0 + \Omega_l^0 g (u^- + u^+) \right\} \mathrm{d}r \\
=& -\omega^2 \int_{\mathcal{R}} \rho r^2 U' U \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \dot{U}' \left[(\kappa + \tfrac{4}{3}\mu)\dot{U} + (\kappa - \tfrac{2}{3}\mu) r^{-1} [2U - \sqrt{2}\Omega_l^0 V] \right] \mathrm{d}r, \\
&+ \int_{\mathcal{R}} r U' \left[\Omega_l^0 \sqrt{2}\mu [\dot{V} - r^{-1}(V - \sqrt{2}\Omega_l^0 U)] + 2(\kappa - \frac{2}{3}\mu)\dot{U} + 2(\kappa + \frac{1}{3}\mu) r^{-1} [2U - \Omega_l^0 \sqrt{2} V] \right] \mathrm{d}r, \\
&+ \int_{\mathcal{R}} r\rho U' \left\{ r\dot{P} + (4\pi G\rho r - 4g) U + \Omega_l^0 g \sqrt{2} V \right\} \mathrm{d}r, \\
=& -\omega^2 \int_{\mathcal{R}} \rho r^2 U' U \,\mathrm{d}r + \kappa \int_{\mathcal{R}} r^2 \dot{U}' \left[\dot{U} + 2r^{-1} U - k r^{-1} V \right] \mathrm{d}r + \mu \int_{\mathcal{R}} r^2 \dot{U}' \left[\frac{4}{3}\dot{U} - \frac{4}{3} r^{-1} U + \frac{2}{3} r^{-1} k V \right] \mathrm{d}r, \\
&+ 2\kappa \int_{\mathcal{R}} r U' [\dot{U} + r^{-1}(2U - kV)] \,\mathrm{d}r + \mu \int_{\mathcal{R}} r U' \left[k[\dot{V} - r^{-1}(V - kU)] - \frac{4}{3}\dot{U} + \frac{4}{3} r^{-1} U - \frac{2}{3} r^{-1} k V \right] \mathrm{d}r, \\
&+ \int_{\mathcal{R}} r\rho U' \left\{ r\dot{P} + (4\pi G\rho r - 4g) U + \Omega_l^0 g \sqrt{2} V \right\} \mathrm{d}r, \\
=& -\omega^2 \int_{\mathcal{R}} \rho r^2 U' U \,\mathrm{d}r + \kappa \int_{\mathcal{R}} r \left[r\dot{U}' + 2U' \right] \left[\dot{U} + 2r^{-1} U - k r^{-1} V \right] \mathrm{d}r, \\
&+ \frac{4}{3}\mu \int_{\mathcal{R}} \left[r^2 \dot{U}' - r U' \right] \left[\dot{U} - r^{-1} U + \frac{1}{2} r^{-1} k V \right] \mathrm{d}r + \mu \int_{\mathcal{R}} r U' k [\dot{V} - r^{-1}(V - kU)] \,\mathrm{d}r, \\
&+ \int_{\mathcal{R}} r\rho U' \left\{ r\dot{P} + (4\pi G\rho r - 4g) U + k g V \right\} \mathrm{d}r. \qquad (232)
\end{aligned}
$$

The $\pm$ component we can combine by setting $w^+ = w^- = V'/\sqrt{2}$. This gives

$$
\begin{aligned}
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \frac{V'}{\sqrt{2}} u^+ \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \frac{\dot{V}'}{\sqrt{2}} T^{0+} \,\mathrm{d}r + \int_{\mathcal{R}} r \frac{V'}{\sqrt{2}} \left\{ \Omega_l^0 \left[T^{-+} + \rho(P + gu^0) \right] + \Omega_l^2 T^{++} - T^{0+} \right\} \mathrm{d}r \\
&-\omega^2 \int_{\mathcal{R}} r^2 \rho \frac{V'}{\sqrt{2}} u^- \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \frac{\dot{V}'}{\sqrt{2}} T^{0-} \,\mathrm{d}r + \int_{\mathcal{R}} r \frac{V'}{\sqrt{2}} \left\{ \Omega_l^0 \left[T^{-+} + \rho(P + gu^0) \right] + \Omega_l^2 T^{--} - T^{0-} \right\} \mathrm{d}r, \\
=& -\omega^2 \int_{\mathcal{R}} r^2 \rho V'(u^+ + u^-)/\sqrt{2} \,\mathrm{d}r + \int_{\mathcal{R}} r^2 \dot{V}'(T^{0+} + T^{0-})/\sqrt{2} \,\mathrm{d}r \\
&+ \sqrt{2}\Omega_l^0 \int_{\mathcal{R}} r V' [T^{-+} + \rho(P + gU)] \,\mathrm{d}r + \int_{\mathcal{R}} r V' \Omega_l^2 (T^{--} + T^{++})/\sqrt{2} \,\mathrm{d}r \\
&- \int_{\mathcal{R}} r V'(T^{0+} + T^{0-})/\sqrt{2} \,\mathrm{d}r, \\
=& -\omega^2 \int_{\mathcal{R}} r^2 \rho V' V \,\mathrm{d}r + \mu \int_{\mathcal{R}} r^2 \dot{V}' [\dot{V} + k r^{-1} U - r^{-1} V] \,\mathrm{d}r \\
&+ k \int_{\mathcal{R}} r V' \left[-(\kappa - \tfrac{2}{3}\mu)\dot{U} - (\kappa + \tfrac{1}{3}\mu) r^{-1} (2U - kV) + \rho(P + gU) \right] \mathrm{d}r \\
&+ 2\mu (\Omega_l^2)^2 \int_{\mathcal{R}} V' V \,\mathrm{d}r - \mu \int_{\mathcal{R}} r V' [\dot{V} + k r^{-1} U - r^{-1} V] \,\mathrm{d}r \\
=& -\omega^2 \int_{\mathcal{R}} r^2 \rho V' V \,\mathrm{d}r + \mu \int_{\mathcal{R}} r^2 \dot{V}' [\dot{V} + k r^{-1} U - r^{-1} V] \,\mathrm{d}r \\
&- \kappa \int_{\mathcal{R}} r (kV') [\dot{U} + 2r^{-1} U - k r^{-1} V] \,\mathrm{d}r + \frac{2}{3}\mu \int_{\mathcal{R}} r (kV') [\dot{U} - r^{-1} U + \frac{1}{2} k r^{-1} V] \,\mathrm{d}r \\
&+ \int_{\mathcal{R}} r \rho (kV')(P + gU) \,\mathrm{d}r + 2\mu (\Omega_l^2)^2 \int_{\mathcal{R}} V' V \,\mathrm{d}r - \mu \int_{\mathcal{R}} r V' [\dot{V} + k r^{-1} U - r^{-1} V] \,\mathrm{d}r, \\
=& -\omega^2 \int_{\mathcal{R}} r^2 \rho V' V \,\mathrm{d}r + \mu \int_{\mathcal{R}} r^2 [\dot{V}' - r^{-1} V'] [\dot{V} + k r^{-1} U - r^{-1} V] \,\mathrm{d}r \\
&- \kappa \int_{\mathcal{R}} r (kV') [\dot{U} + 2r^{-1} U - k r^{-1} V] \,\mathrm{d}r + \frac{2}{3}\mu \int_{\mathcal{R}} r (kV') [\dot{U} - r^{-1} U + \frac{1}{2} r^{-1} k V] \,\mathrm{d}r \\
&+ \int_{\mathcal{R}} r \rho (kV')(P + gU) \,\mathrm{d}r + 2\mu (\Omega_l^2)^2 \int_{\mathcal{R}} V' V \,\mathrm{d}r.
\end{aligned} \tag{233}
$$

Finally the equation for gravity is

$$
\int_{\mathcal{R}} \left[r^2 \dot{P}' \dot{P} + k^2 P' P \right] \mathrm{d}r + b(l+1) P' P = -4\pi G \int_{\mathcal{R}} \rho \left[r^2 \dot{P}' U + k r P' V \right] \mathrm{d}r. \tag{234}
$$

We now combine (232), (233) and (234) and use $2(\Omega_l^2)^2 = (l+2)(l-1) = l^2+l-1 = l(l+1)-2 = k^2-2$, we have

$$
\begin{aligned}
= & -\omega^2 \int_{\mathcal{R}} \rho r^2 U'U \,\mathrm{d}r + \kappa \int_{\mathcal{R}} r \left[r\dot{U}' + 2U'\right] \left[\dot{U} + 2r^{-1}U - kr^{-1}V\right] \mathrm{d}r \\
& + \frac{4}{3}\mu \int_{\mathcal{R}} \left[r^2\dot{U}' - rU'\right] \left[\dot{U} - r^{-1}U + \frac{1}{2}kr^{-1}V\right] \mathrm{d}r + \mu \int_{\mathcal{R}} rU'k[\dot{V} + kr^{-1}U - r^{-1}V] \,\mathrm{d}r \\
& + \int_{\mathcal{R}} r\rho U' \left\{r\dot{P} + (4\pi G\rho r - 4g)U + kgV\right\} \mathrm{d}r \\
& - \omega^2 \int_{\mathcal{R}} r^2 \rho V'V \,\mathrm{d}r + \mu \int_{\mathcal{R}} r^2[\dot{V}' - r^{-1}V'][\dot{V} + kr^{-1}U - r^{-1}V] \,\mathrm{d}r \\
& - \kappa \int_{\mathcal{R}} r(kV')[\dot{U} + 2r^{-1}U - kr^{-1}V] \,\mathrm{d}r + \frac{2}{3}\mu \int_{\mathcal{R}} r(kV') \left[\dot{U} - r^{-1}U + \frac{1}{2}kr^{-1}V\right] \mathrm{d}r \\
& + \int_{\mathcal{R}} r\rho(kV')(P + gU) \,\mathrm{d}r + 2\mu(\Omega_l^2)^2 \int_{\mathcal{R}} V'V \,\mathrm{d}r \\
& + \frac{1}{4\pi G} \int_{\mathcal{R}} \left[r^2\dot{P}'\dot{P} + k^2P'P\right] \mathrm{d}r + \frac{b(l+1)}{4\pi G} P'P + \int_{\mathcal{R}} \rho \left[r^2\dot{P}'U + krP'V\right] \mathrm{d}r
\end{aligned} \tag{235}
$$

Further simplification gives

$$
\begin{aligned}
& -\omega^2 \int_{\mathcal{R}} \rho r^2[U'U + V'V] \,\mathrm{d}r + (k^2-2) \int_{\mathcal{R}} \mu V'V \,\mathrm{d}r + \int_{\mathcal{R}} 4\rho(\pi G\rho r - g)U'Ur \,\mathrm{d}r \\
& + \int_{\mathcal{R}} \kappa \left[r\dot{U}' + 2U' - kV'\right] \left[r\dot{U} + 2U - kV\right] \mathrm{d}r \\
& + \frac{4}{3} \int_{\mathcal{R}} \mu \left[r\dot{U}' - U' + \frac{1}{2}kV'\right] \left[r\dot{U} - U + \frac{1}{2}kV\right] \mathrm{d}r \\
& + \int_{\mathcal{R}} \mu[r\dot{V}' + kU' - V'][r\dot{V} + kU - V] \,\mathrm{d}r \\
& + \frac{1}{4\pi G} \int_{\mathcal{R}} \left[r^2\dot{P}'\dot{P} + k^2P'P\right] \mathrm{d}r + \frac{b(l+1)}{4\pi G} P'P \\
& + \int_{\mathcal{R}} \rho[U'\dot{P} + \dot{P}'U]r^2 \,\mathrm{d}r + k \int_{\mathcal{R}} \rho[P'V + V'P]r \,\mathrm{d}r + k \int_{\mathcal{R}} \rho g[U'V + UV']r \,\mathrm{d}r.
\end{aligned} \tag{236}
$$

Traditionally, the tangential component is redefined, compared to what we have used, as $V \to kV$, i.e. $kV = (u^- + u^+)/\sqrt{2}$, giving

$$
\begin{aligned}
&- \omega^2 \int_{\mathcal{R}} \rho r^2 [U'U + k^2 V'V] \,\mathrm{d}r + k^2(k^2-2) \int_{\mathcal{R}} \mu V'V \,\mathrm{d}r + \int_{\mathcal{R}} 4\rho(\pi G \rho r - g) U'U r \,\mathrm{d}r \\
&+ \int_{\mathcal{R}} \kappa \left[r\dot{U}' + 2U' - k^2 V' \right] \left[r\dot{U} + 2U - k^2 V \right] \mathrm{d}r \\
&+ \frac{4}{3} \int_{\mathcal{R}} \mu \left[r\dot{U}' - U' + \frac{1}{2} k^2 V' \right] \left[r\dot{U} - U + \frac{1}{2} k^2 V \right] \mathrm{d}r \\
&+ k^2 \int_{\mathcal{R}} \mu [r\dot{V}' + U' - V'][r\dot{V} + U - V] \,\mathrm{d}r \\
&+ \frac{1}{4\pi G} \int_{\mathcal{R}} \left[r^2 \dot{P}'\dot{P} + k^2 P'P \right] \mathrm{d}r + \frac{b(l+1)}{4\pi G} P'P \\
&+ \int_{\mathcal{R}} \rho [U'\dot{P} + \dot{P}'U] r^2 \,\mathrm{d}r + k^2 \int_{\mathcal{R}} \rho [P'V + V'P] r \,\mathrm{d}r + k^2 \int_{\mathcal{R}} \rho g [U'V + UV'] r \,\mathrm{d}r. \qquad (237)
\end{aligned}
$$

Force term

$$
\begin{aligned}
&H(\omega) \Bigg\{ M^{00} \dot{U}' \overline{Y^0(\theta_s, \varphi_s)} + \\
&+ r_s^{-1} U'(r_s) \left\{ \frac{k}{\sqrt{2}} \left[M^{+0} \overline{Y^+(\theta_s, \varphi_s)} + M^{-0} \overline{Y^-(\theta_s, \varphi_s)} \right] - \overline{Y^0(\theta_s, \varphi_s)} \left[M^{+-} + M^{-+} \right] \right\} \\
&+ \frac{k}{\sqrt{2}} \dot{V}'(r_s) \left[M^{0+} \overline{Y^+(\theta_s, \varphi_s)} + M^{0-} \overline{Y^-(\theta_s, \varphi_s)} \right] + \\
&+ \frac{k}{\sqrt{2}} r_s^{-1} V'(r_s) \Big[\frac{k}{\sqrt{2}} \left(M^{-+} + M^{+-} \right) \overline{Y^0(\theta_s, \varphi_s)} - \left(M^{+0} \overline{Y^+(\theta_s, \varphi_s)} + M^{-0} \overline{Y^-(\theta_s, \varphi_s)} \right) \\
&\qquad + \Omega_l^2 \left(M^{++} \overline{Y^{++}(\theta_s, \varphi_s)} + M^{--} \overline{Y^{--}(\theta_s, \varphi_s)} \right) \Big] \Bigg\}. \qquad (238)
\end{aligned}
$$

8.7.2 Numerical approach

The weak form that we have obtained includes both tangential and radial displacements, as well as gravity. Therefore, we need to include all of these terms within the matrix formulation that we will use for the spectral element solution. The best way to store the components is still up to consideration. However, for memory access purposes it may be best to have coefficients that are at the same radius close in memory. In other words we store U, V, P at each radius. In this manner it will hopefully be well set up for memory access.

To deal with the fluid-solid boundary we need to see what works. What I will probably do, however, is take the approach of Kemper et al. They have allowed the degree of freedom on the FS boundary to be free on both sides for V.

Potential issues:

1. At zero radius we need $U = V = 0$. However, P can be non-zero. Can simply remove the point at 0? Definitely best to have a local to global map function rather than iterating for this problem. We can associate U, V and P as the 0-th, 1-st and 2-nd eigenfunctions and maybe have separate functions for finding indices as well.

2. Fluid-solid interface. At the FS interface you can have slip of the fluid against the solid. The tangential traction needs to be zero, this should be enforced by the weak form though I think.

Particulars of the local to global map:

1. We label $U \equiv 0$, $V \equiv 1$ and $P \equiv 2$ in terms of the numbering of eigenfunctions.
2. U and V have to be zero at 0 so we don't include those points
3. U and P must be continuous on FS interfaces, but we need to allow V to have separate DOFs at the two sides of the interface

Ordering:

$$\begin{bmatrix} P_{0,0} \\ U_{0,1} \\ V_{0,1} \\ P_{0,1} \\ U_{0,2} \\ \vdots \\ U_{N_{FS},N_N} \\ V_{N_{FS},N_N} \\ P_{N_{FS},N_N} \\ V_{N_{FS}+1,0} \\ U_{N_{FS}+1,1} \\ V_{N_{FS}+1,1} \\ P_{N_{FS}+1,0} \\ \vdots \end{bmatrix} \tag{239}$$

Local-to-global map:

1. Need to check if in first element, in which case, if trying to find U or V then throw an error, if P return 0.
2. Check if element number is equal to or lower than the first FS boundary
3. Store a vector that holds the offset at each element? This makes some sense
4. If element is on the upper side of a FS boundary then check if we are returning for V. If that's the case then give the value on the lower end of the element, else return the shared value

8.7.3 Forming the matrices

The mass matrix is given by

$$M = \int_{\mathcal{R}} \rho r^2 [U'U + k^2 V'V] \, \mathrm{d}r. \tag{240}$$

This is diagonal. As we use a spectral element method we need to simply multiply the value of ρr^2 by the weight at the point we are considering. The $U'U$ and $V'V$ terms are separate.

The kinetic energy matrix is given by

$$
\begin{aligned}
K = {} & k^2(k^2-2)\int_{\mathcal{R}} \mu V'V \,\mathrm{d}r + \int_{\mathcal{R}} 4\rho(\pi G\rho r - g)U'Ur \,\mathrm{d}r \\
& + \int_{\mathcal{R}} \kappa \left[r\dot{U}' + 2U' - k^2V'\right]\left[r\dot{U} + 2U - k^2V\right] \mathrm{d}r \\
& + \frac{4}{3}\int_{\mathcal{R}} \mu \left[r\dot{U}' - U' + \frac{1}{2}k^2V'\right]\left[r\dot{U} - U + \frac{1}{2}k^2V\right] \mathrm{d}r \\
& + k^2\int_{\mathcal{R}} \mu[r\dot{V}' + U' - V'][r\dot{V} + U - V] \,\mathrm{d}r \\
& + \frac{1}{4\pi G}\int_{\mathcal{R}} \left[r^2\dot{P}'\dot{P} + k^2P'P\right] \mathrm{d}r + \frac{b(l+1)}{4\pi G}P'P \\
& + \int_{\mathcal{R}} \rho[U'\dot{P} + \dot{P}'U]r^2 \,\mathrm{d}r + k^2\int_{\mathcal{R}} \rho[P'V + V'P]r \,\mathrm{d}r + k^2\int_{\mathcal{R}} \rho g[U'V + UV']r \,\mathrm{d}r. \qquad (241)
\end{aligned}
$$

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